GOOGLE WORKSPACE

CHEAT SHEETS

Drive, Docs, Sheets, Slides & Gmail — Shortcuts, Formulas and Apps Script

Drive Search · Sharing · Version History · Smart Chips · Templates · Styles · Document Tabs · Suggesting · Tables · Filter Views · XLOOKUP · ARRAYFORMULA · QUERY · REGEX · IMPORTRANGE · LAMBDA · Pivot Tables · Slides Themes · Presenter View · Gmail Filters · Calendar · Forms · Macros · Apps Script · Gemini

32 REFERENCE SHEETS

CONTENTS

DRIVE & SHARED BASICS

01 Introduction to Google Workspace
02 Drive Organization & Search
03 Sharing, Comments & Version History
04 Templates & Smart Chips
05 Shortcuts That Work Everywhere

DOCS

06 Styles & Formatting
07 Long Documents & Document Tabs
08 Writing Faster
09 Review, Mail Merge & Export

SHEETS

10 Sheets Navigation & Selection
11 Data Entry, Validation & Cleanup
12 Formula Basics
13 Tables & Filter Views
14 Logic & Math Functions
15 Lookup Functions
16 Array Formulas
17 The QUERY Function
18 Text, Date & Regex Functions
19 Import & Live Data
20 LET, LAMBDA & Named Functions
21 Pivot Tables, Charts & Conditional Formatting

SLIDES

22 Themes, Master & Building Slides
23 Presenting

GMAIL, CALENDAR & FORMS

24 Gmail Efficiency
25 Calendar & Meet
26 Forms, Keep & Tasks

AUTOMATION & AI

27 Macros & Apps Script Basics
28 Apps Script Recipes
29 Gemini in Workspace

QUICK REFERENCE

30 Quick Ref — Drive, Docs, Slides & Gmail
31 Quick Ref — Sheets Shortcuts
32 Quick Ref — Sheets Formulas

GWS

Table of Contents

32 sheets · syntax · examples · notes

DRIVE & SHARED BASICS 01 – 05

- 01 Introduction to Google…
- 02 Drive Organization & S…
- 03 Sharing, Comments & …
- 04 Templates & Smart Chi…
- 05 Shortcuts That Work E…

DOCS 06 – 09

- 06 Styles & Formatting
- 07 Long Documents & Do…
- 08 Writing Faster
- 09 Review, Mail Merge & E…

SHEETS 10 – 21

- 10 Sheets Navigation & S…
- 11 Data Entry, Validation …
- 12 Formula Basics
- 13 Tables & Filter Views
- 14 Logic & Math Functions
- 15 Lookup Functions
- 16 Array Formulas
- 17 The QUERY Function
- 18 Text, Date & Regex Fun…
- 19 Import & Live Data
- 20 LET, LAMBDA & Named…
- 21 Pivot Tables, Charts & …

SLIDES 22 – 23

- 22 Themes, Master & Buil…
- 23 Presenting

GMAIL, CALENDAR & FORMS 24 – 26

- 24 Gmail Efficiency
- 25 Calendar & Meet
- 26 Forms, Keep & Tasks

AUTOMATION & AI 27 – 29

- 27 Macros & Apps Script …
- 28 Apps Script Recipes
- 29 Gemini in Workspace

QUICK REFERENCE 30 – 32

- 30 Quick Ref — Drive, Doc…
- 31 Quick Ref — Sheets Sh…
- 32 Quick Ref — Sheets Fo…

Sheet 01

Introduction to Google Workspace

free account vs Workspace · docs.new / sheets.new / slides.new · offline mode · Drive for desktop · mobile apps · Office files

EVERYTHING LIVES IN THE BROWSER

Google Workspace is Docs, Sheets, Slides, Gmail, Drive, Calendar, Meet and friends. Every file lives in **Drive**, saves as you type, and can be opened by anyone you share it with. A free Google account gets the same editors; paid **Workspace** plans add custom domains, more storage, admin controls, and extras like eSignature, appointment booking pages and Gemini features.

NOTES

- Native Google files are links, not documents on disk; download to get `.docx` / `.xlsx` / `.pptx`
- Office files can be edited directly ("Office editing" mode) or converted with File › Save as Google Docs
- **Offline**: Drive settings › Offline, then mark files Available offline (Chrome or Edge)
- **Drive for desktop** shows Drive as a disk in Finder / File Explorer
- Chrome is the best-supported browser; most features work in Safari, Edge and Firefox

OPEN THINGS FAST

```
docs.new    sheets.new    slides.new      # new blank file in My Drive
forms.new   sites.new     keep.new        # other apps
drive.google.com/drive/recent             # recently opened files

⌘/            / Ctrl+/          # show all shortcuts (any editor)
⌥/            / Alt+/           # search the menus
⌘S            / Ctrl+S          # nothing to save — it already did
⌘O            / Ctrl+O          # open a file from Drive
⌘P            / Ctrl+P          # print / PDF
File › Download › .docx / .xlsx / .pptx / .pdf
```

CODE EXPLAINED

- → The `.new` shortcuts work in any browser address bar while signed in
- → Shortcuts are listed **macOS first**, then **Windows / ChromeOS**, on every sheet
- → Browser shortcuts can win: `⌘N` opens a browser window, not a document
- → Version history replaces "Save As" for keeping old copies (sheet 03)

OTHER SUITES

- ⇄ **Microsoft 365** — `word.new`, `excel.new`; desktop apps plus web apps in OneDrive
- ⇄ **Apple iWork** — Pages / Numbers / Keynote on Mac / iPad; iCloud.com for web versions

PLATFORM DIFFERENCES

- ▣ **Browser** — full feature set
- ▣ **Offline / Drive for desktop** — edit marked files without internet; syncs later
- ▣ **Mobile** — Docs, Sheets, Slides apps; fewer menus, no add-ons or Apps Script editor

ACCOUNT TYPES

ACCOUNT	GETS
Free Google account	15 GB shared storage, all editors
Google One	More storage, some AI features
Workspace Business	Custom domain, admin, shared drives
Workspace Enterprise	Security, compliance, more Gemini
Education / nonprofit	Discounted Workspace editions

APP MAP

APP	REPLACES
Docs	Word / Pages
Sheets	Excel / Numbers
Slides	PowerPoint / Keynote
Gmail + Calendar	Outlook / Apple Mail
Drive	OneDrive / iCloud Drive
Meet + Chat	Teams / FaceTime

⚠ COMMON MISTAKES

- Downloading a Google file to "save" it; the Drive copy is already saved and current
- Emailing a `.gdoc` shortcut file from Drive for desktop; it's only a link, not the document
- Expecting complex Word or Excel features (macros, Power Query) to survive conversion

✓ TIPS & BEST PRACTICES

- Bookmark `docs.new` / `sheets.new` or type them; fastest way to a blank file
- Turn on offline mode before travelling
- Keep Office files as Office files if they go back to Office users; edit them in Office mode

QUICK SUMMARY

- `*.new` — instant blank file
- `⌘/` / `Ctrl+/` — shortcut list
- `⌥/` / `Alt+/` — search menus
- `File › Download` — Office / PDF copies

Sheet 02

Drive Organization & Search

My Drive vs Shared drives · Drive shortcuts · search operators · Priority & Activity · folder colours · storage cleanup

FIND FILES INSTEAD OF FILING THEM

Drive search is fast and supports **operators** like `type:`, `owner:` and `before:`, so you rarely need deep folder trees. **Shared drives** (Workspace) belong to the team, not a person, so files survive when people leave. A **shortcut** (⇧Z) puts the same file in several folders without copying it.

NOTES

- `/` jumps to the Drive search box; click the sliders icon for advanced search
- **Shared with me** only shows files; add them to My Drive with a shortcut
- Shortcuts (`⇧Z`) are pointers; deleting a shortcut doesn't delete the file
- Star important files (`S`); **Starred** works like a favourites list
- Storage: drive.google.com/drive/quota lists the largest files

DRIVE SEARCH & KEYS

```
type:spreadsheet owner:me before:2026-01-01   # my old sheets
type:pdf "invoice" after:2026-09-01           # recent PDF invoices
owner:dana@acme.com is:starred                # Dana's starred files
title:"Q3 plan"                               # name only
to:me  /  from:me                             # shared with / by me

/              # focus search
⇧Z             # add shortcut to another folder
S              # star / unstar
.              # share selected file
N  /  ⇧F       # rename selected / new folder
⇧T  ⇧S  ⇧P     # new Doc / Sheet / Slides from Drive
```

CODE EXPLAINED

- → Operators combine with spaces (AND); `OR` must be capitalised
- → Dates use `YYYY-MM-DD`
- → Drive keyboard shortcuts work when the file list has focus; `?` shows them all
- → Searching for words inside PDFs and images works because Drive runs OCR

OTHER SUITES

- ⇄ **Microsoft 365** — OneDrive / SharePoint search, Add shortcut to My files, file Shared view
- ⇄ **Apple iWork** — iCloud Drive in Finder; Spotlight search and Finder tags

PLATFORM DIFFERENCES

- ▣ **Browser** — operators, Activity panel, shortcuts
- ▣ **Drive for desktop** — Finder / Explorer search, Spotlight indexes file names
- ▣ **Mobile** — search with filters; fewer operators

SEARCH OPERATORS

OPERATOR	FINDS
`type:document / spreadsheet / pdf / image`	File type
`owner:me / owner:email`	Who owns it
`before: / after:YYYY-MM-DD`	Modified date
`is:starred / is:trashed`	State
`title:"words"`	File name only
`to: / from:`	Shared to / by

MY DRIVE VS SHARED DRIVE

	MY DRIVE	SHARED DRIVE
`Owned by`	You	The team
`When you leave`	Files transfer / deleted	Stay put
`Membership`	Per file	Whole drive
`Available on`	All accounts	Workspace plans

⚠ COMMON MISTAKES

- Making copies of a file for each folder; use a shortcut so there's one version
- Keeping team files in your own My Drive, so they disappear when you leave
- Hunting through Shared with me instead of searching by owner

✓ TIPS & BEST PRACTICES

- Colour key folders (right-click › Organize › Folder colour)
- Use the Activity panel (`i` details pane) to see who changed what
- Empty Trash and sort by storage to free space

QUICK SUMMARY

- `type: owner: before:` — search operators
- `⇧Z` — shortcut, not copy
- `Shared drives` — team-owned files
- `/` — jump to search

Sheet 03

Sharing, Comments & Version History

Viewer / Commenter / Editor · link sharing · expiring access · action items · version history · named versions · Approvals

ONE FILE, MANY PEOPLE

Share a file with people or a link and choose **Viewer**, **Commenter** or **Editor**. Comments can **assign action items** (+name), which email the person and show up in their task list. **Version history** keeps every change: name important versions, compare, and restore. **Approvals** (Workspace) collect sign-off on a final version.

NOTES

- Share: the **Share** button (top right) in any editor or `.` in Drive
- Link sharing: Restricted, your organization, or Anyone with the link
- Expiring access (Workspace): set an end date per person
- Comment: `⌘⌥M` / `Ctrl+Alt+M`; type `+email` to assign it
- Version history: `⌘⌥⇧H` / `Ctrl+Alt+Shift+H`

SHARE AND TRACK CHANGES

1. Click **Share**, add people, pick Viewer / Commenter / Editor
2. Under **General access**, keep **Restricted** unless a link must work for others
3. Settings (gear): stop editors changing permissions; stop viewers downloading
4. Reviewers comment or use **Suggesting** mode (Docs, sheet 09)
5. Before big edits: File › Version history › **Name current version**
6. To roll back: open Version history, pick a version, **Restore this version**

WHERE TO FIND IT

TASK	WHERE / SHORTCUT
Share	Share button / . in Drive
Comment	⌘⌥M / Ctrl+Alt+M
Assign action item	Comment › +name
Version history	⌘⌥⇧H / Ctrl+Alt+Shift+H
Name a version	File › Version history › Name
Approvals	File › Approvals (Workspace)
Transfer ownership	Share › person › Transfer

OTHER SUITES

⇄ **Microsoft 365** — Share in OneDrive, @mention comments, File › Info › Version History

⇄ **Apple iWork** — iCloud Collaborate, comments, Browse All Versions

PLATFORM DIFFERENCES

- **Browser** — every sharing option
- **Offline** — edits sync later; sharing needs a connection
- **Mobile** — share, comment, view version history in Docs / Sheets

ROLES

ROLE	CAN
Viewer	Read, download (unless blocked)
Commenter	Comment and suggest
Editor	Change content, share (by default)
Owner	Delete, transfer ownership
Shared drive Manager	Manage members and files

LINK TRICKS

CHANGE URL ENDING TO	GIVES
/copy	Prompt to make a copy
/template/preview	Template-style preview + Use template
/preview	Clean read-only view
/export?format=pdf	Direct PDF download
/edit#gid=123 (Sheets)	Opens a specific tab

⚠ COMMON MISTAKES

- Sharing with "Anyone with the link" for internal documents
- Making copies to "freeze" a version instead of naming a version
- Leaving editors able to reshare sensitive files; change it in Share settings

✓ TIPS & BEST PRACTICES

- Send `/copy` links for templates so people never edit your master
- Assign comments with `+name` instead of emailing about them
- Use Version history's **Show changes** to see who edited what

QUICK SUMMARY

- `Viewer / Commenter / Editor` — roles
- `+name` — assign a comment
- `Name current version` — save milestones
- `/copy` — force a copy

Templates & Smart Chips

START FROM STRUCTURE

Docs, Sheets and Slides each have a **template gallery**; Workspace domains can add their own templates. Inside documents, type @ to insert **smart chips**: people, files, dates, places, dropdowns and **building blocks** like meeting notes, project trackers and email drafts. Chips are live: hover a person for their contact card, a file for a preview.

NOTES

- Gallery: Docs home › Template gallery (or File › New › From template)
- Custom templates (Workspace): Template gallery › Submit template
- Free accounts: share a master with a `/copy` link (sheet 03)
- `@` menu: people, files, calendar events, dates, dropdowns, building blocks
- Dropdown chips (status, priority) can be customised and reused

AUTOMATE IT

- A `/copy` link to your master file means people always start from the template
- Building blocks (`@meeting notes`, `@project roadmap`) instead of rebuilding layouts

MAKE AND USE A TEMPLATE

1. Build the file with styles, headings, placeholders and dropdown chips
2. Free account: Share › Copy link, change `/edit…` to `/copy` and send that
3. Workspace: Template gallery › your org tab › **Submit template**
4. Inside docs, type `@` and pick a building block (e.g. Meeting notes)
5. Type `@date` or `@today` for date chips; `@` + name for people chips
6. Insert › Dropdown to create a status chip, then copy-paste it down

WHERE TO FIND IT

TASK	WHERE
`Template gallery`	Docs / Sheets / Slides home
`Smart chip menu`	Type @
`Dropdown chip`	Insert › Dropdown
`Building blocks`	Insert › Building blocks
`Convert link to chip`	Paste link › Tab
`Submit org template`	Template gallery › Submit

OTHER SUITES

- ⇄ **Microsoft 365** — File › New templates; Loop components for live tables and checklists
- ⇄ **Apple iWork** — template chooser, Save as Template; no live chips

PLATFORM DIFFERENCES

- **Browser** — all chips and building blocks
- **Offline** — chips show, previews need a connection
- **Mobile** — insert people and date chips; fewer building blocks

SMART CHIPS

TYPE @…	INSERTS
`@name`	Person chip with contact card
`@file name`	File chip with preview
`@today / @date`	Date chip with picker
`@meeting notes`	Notes block tied to an event
`@dropdown`	Status / priority dropdown
`@place`	Google Maps place chip

BUILDING BLOCKS

BLOCK	GIVES
`Meeting notes`	Attendees, agenda, action items
`Email draft`	To / subject / body, sends via Gmail
`Project roadmap`	Table with status chips
`Review tracker`	Reviewer, status, notes
`Content tracker`	Writer, due date, status

⚠ COMMON MISTAKES

- Editing the shared master because people opened it instead of copying
- Typing dates as text so they don't sort or remind; use date chips
- Creating a new dropdown per row instead of copying one chip

✓ TIPS & BEST PRACTICES

- Use the `/template/preview` link for a friendlier "Use template" page
- Put status dropdowns in tables to make lightweight trackers in Docs
- Email draft blocks let a team write an email together before sending

QUICK SUMMARY

- `Template gallery` — start files
- `@` — smart chips menu
- `/copy` — template link trick
- `Building blocks` — ready-made sections

Sheet 05

Shortcuts That Work Everywhere

⌘/ / Ctrl+/ shortcut list · ⌥/ / Alt+/ search menus · ⌘⇧V paste plain · paint format · voice typing ⌘⇧S · Gemini panel

LEARN THESE IN EVERY EDITOR

Docs, Sheets and Slides share core shortcuts. The two to learn first: ⌘/ / Ctrl+/ shows every shortcut, and ⌥/ / Alt+/ **searches the menus** so you can run any command by typing its name. **Paste without formatting** and **paint format** keep files tidy; **voice typing** and the **Gemini** side panel help you draft.

NOTES

- Search the menus also finds functions in Sheets and settings like "word count"
- Paint format: click the roller, then the target; double-click to keep it on
- Voice typing (Docs, Slides speaker notes): Tools › Voice typing
- Enable compatible shortcuts in the shortcut list to get extra keys (e.g. Excel-style in Sheets)
- Browser shortcuts (`⌘T` , `⌘W`) still belong to the browser

SHARED EDITOR SHORTCUTS

```
⌘/              / Ctrl+/              # keyboard shortcut list
⌥/              / Alt+/               # search the menus
⌘C ⌘X ⌘V        / Ctrl+C X V          # copy, cut, paste
⌘⇧V             / Ctrl+Shift+V        # paste without formatting
⌘⌥C  ⌘⌥V        / Ctrl+Alt+C  V       # copy / paste formatting
⌘Z  ⌘⇧Z         / Ctrl+Z  Ctrl+Y      # undo / redo
⌘K              / Ctrl+K              # insert link
⌘F  ⌘⇧H         / Ctrl+F  Ctrl+H      # find / find & replace
⌘⌥M             / Ctrl+Alt+M          # comment
⌘⇧S             / Ctrl+Shift+S        # voice typing (Docs)
⌘\              / Ctrl+\              # clear formatting
```

CODE EXPLAINED

- → `⌘⌥C` / `⌘⌥V` is the keyboard version of paint format
- → Search the menus is the closest thing to Office's Tell Me / KeyTips
- → Clear formatting removes character formatting but keeps paragraph styles
- → Compatible shortcuts are toggled at the bottom of the `⌘/` dialog

OTHER SUITES

- ⇄ **Microsoft 365** — Paste Special `^⌘V` / `Ctrl+Alt+V`, Format Painter, Search box `Alt+Q`
- ⇄ **Apple iWork** — Paste and Match Style `⌥⇧⌘V`, Copy / Paste Style `⌥⌘C` / `⌥⌘V`

PLATFORM DIFFERENCES

- ▣ **Browser** — every shortcut
- ▣ **Offline** — same shortcuts
- ▣ **Mobile** — no menu search; long-press for paste options

MOVING & SELECTING (DOCS / SLIDES TEXT)

ACTION	MAC / WINDOWS
Word left / right	⌥← → / Ctrl+← →
Line start / end	⌘← → / Home End
Doc start / end	⌘↑ ↓ / Ctrl+Home End
Select word by word	⌥⇧← → / Ctrl+Shift+← →
Select all	⌘A / Ctrl+A
Select all matching text (Docs)	right-click › Select all matching

GETTING HELP

NEED	DO
Any command	⌥/ and type
A shortcut	⌘/ and search
Draft text	Gemini side panel (✦ icon)
Spell check	⌘⌥X / Ctrl+Alt+X
Word count	⌘⇧C / Ctrl+Shift+C

⚠ COMMON MISTAKES

- Pasting from a website with all its fonts; use `⌘⇧V` / `Ctrl+Shift+V`
- Clicking through menus for rarely used commands; type them into menu search
- Expecting Office shortcuts in Sheets without turning on compatible shortcuts

✓ TIPS & BEST PRACTICES

- Make `⌥/` / `Alt+/` your first move for anything you can't find
- Double-click paint format to apply it to several places
- Check the shortcut dialog every few weeks and learn one new key

QUICK SUMMARY

- `⌘/` — shortcut list
- `⌥/` — search menus
- `⌘⇧V` — paste plain
- `⌘⌥C / V` — copy / paste format

Styles & Formatting

HEADINGS MAKE EVERYTHING ELSE WORK

Docs has **paragraph styles**: Normal text, Title, Subtitle and Heading 1–6. Use them instead of bold and big fonts and you get a document outline, a clickable table of contents and consistent formatting. Change a style once with **Update heading to match** and every heading follows; **Save as my default styles** applies it to new documents.

NOTES

- Styles dropdown is on the toolbar; shortcuts `⌘⌥0` (Normal) to `⌘⌥6`
- Format › Paragraph styles › Heading 1 › **Update 'Heading 1' to match**
- Options › **Save as my default styles** makes your look the default
- **Pageless** (File › Page setup) removes page breaks for on-screen docs
- Line & paragraph spacing: Format › Line & paragraph spacing › Custom

FORMATTING SHORTCUTS

```
⌘⌥0           / Ctrl+Alt+0        # Normal text
⌘⌥1 … ⌘⌥6     / Ctrl+Alt+1 … 6    # Heading 1-6
⌘\            / Ctrl+\            # clear formatting
⌘B ⌘I ⌘U      / Ctrl+B I U        # bold, italic, underline
⌘⇧X           / Alt+Shift+5       # strikethrough
⌘. ⌘,         / Ctrl+. Ctrl+,     # superscript / subscript
⌘⇧> ⌘⇧<       / Ctrl+Shift+> <    # bigger / smaller font
⌘⇧L E R J     / Ctrl+Shift+L E R J  # left, center, right, justify
⌘⇧7 ⌘⇧8       / Ctrl+Shift+7 8    # numbered / bulleted list
⌘] ⌘[         / Ctrl+] Ctrl+[     # indent / outdent
```

CODE EXPLAINED

- → Heading shortcuts both style the text and add it to the outline
- → Clear formatting resets to the paragraph style, so a messy paste becomes clean Normal text
- → Custom fonts: font menu › More fonts (any Google Font)
- → Docs has no character styles; paint format fills that gap

OTHER SUITES

- ⇄ **Microsoft 365** — Word Styles gallery, Update to Match Selection, `⌥⌘1` / `Ctrl+Alt+1` headings
- ⇄ **Apple iWork** — Pages paragraph styles, Update button in the Format sidebar

PLATFORM DIFFERENCES

- ▣ **Browser** — every option
- ▣ **Offline** — same
- ▣ **Mobile** — apply styles; can't update style definitions

PARAGRAPH STYLES

STYLE	USE
`Title / Subtitle`	Document title (not in TOC)
`Heading 1`	Main sections
`Heading 2–3`	Subsections
`Heading 4–6`	Rarely needed
`Normal text`	Body copy

PAGE SETUP

SETTING	WHERE
`Pages vs pageless`	File › Page setup
`Margins / size / orientation`	File › Page setup
`Page colour`	File › Page setup
`Section break`	Insert › Break › Section break
`Columns`	Format › Columns

⚠ COMMON MISTAKES

- Bold Normal text as headings; nothing appears in the outline or TOC
- Fixing every heading by hand after changing your mind; update the style once
- Using Enter for spacing instead of paragraph spacing

✓ TIPS & BEST PRACTICES

- Paste messy text, select it, press `⌘\` / `Ctrl+\`, then apply styles
- Set your fonts once and Save as my default styles
- Use pageless for wide tables and on-screen documents

QUICK SUMMARY

- `⌘⌥1–6` — headings
- `⌘⌥0` — Normal text
- `Update to match` — change a style
- `⌘\` — clear formatting

Sheet 07

Long Documents & Document Tabs

document outline · document tabs · table of contents · headers / footers · different first page · section breaks · footnotes · bookmarks

NAVIGATE AND STRUCTURE BIG DOCS

The **document outline** (left panel) lists every heading for quick jumps. **Document tabs** split one file into named tabs and sub-tabs, like chapters, each with its own outline. Insert a **table of contents** that links to headings, use **section breaks** for different headers or orientation, and add **footnotes** and **bookmarks** for references.

NOTES

- Show outline / tabs: View › Show tabs & outlines, or the icon at top left
- Add a tab: the + next to Document tabs; drag to nest as sub-tabs
- TOC: Insert › Table of contents (with page numbers or with links); refresh with the ↻ button
- Headers: Insert › Headers & footers; **Different first page** in Options
- Footnote: `⌘⌥F` / `Ctrl+Alt+F`

AUTOMATE IT

- A template with tabs already set up (Overview, Requirements, Decisions…) for every project
- Link to headings (Insert › Link › Headings) instead of writing "see page 12"

LONG-DOC KEYS

```
⌘↩            / Ctrl+Enter          # page break
⌘⌥F           / Ctrl+Alt+F          # footnote
^⌘ N then H   / Ctrl+Alt N then H     # next heading
^⌘ P then H   / Ctrl+Alt P then H     # previous heading
^⌘ N then F   / Ctrl+Alt N then F     # next footnote
⌘⌥A ⌘⌥H       / Ctrl+Alt+A Ctrl+Alt+H # show document outline
Insert › Bookmark · Insert › Link › Bookmarks / Headings
Insert › Break › Section break (next page / continuous)
```

CODE EXPLAINED

- → Heading-to-heading jumps are two-key sequences: hold the modifiers, then press N (or P) then H
- → Each document tab has its own outline and can be linked to directly
- → The TOC is not live; click its refresh icon after adding headings
- → Section breaks let headers, margins and orientation change mid-document

OTHER SUITES

- ⇄ **Microsoft 365** — Word Navigation pane, TOC, captions & cross-references, section breaks
- ⇄ **Apple iWork** — Pages TOC sidebar, section breaks, footnotes; no tabs

PLATFORM DIFFERENCES

- ▣ **Browser** — tabs, outline, TOC, bookmarks
- ▣ **Offline** — same
- ▣ **Mobile** — view tabs and outline; limited header / section editing

TABS VS SEPARATE DOCS

USE TABS WHEN	USE SEPARATE DOCS WHEN
Same audience and owner	Different sharing needed
Chapters of one thing	Unrelated topics
You want one link	Each needs its own history
Under ~100 pages	Huge content (performance)

HEADER & FOOTER OPTIONS

OPTION	EFFECT
Different first page	Blank header on cover
Different odd & even	Book-style layout
Link to previous (off)	New header per section
Page numbers	Insert › Page numbers
Start at	Page numbers › Options

⚠ COMMON MISTAKES

- Forgetting to refresh the TOC before sharing or exporting
- Splitting one project into five docs when tabs would keep one link
- Typing page references; link to headings or bookmarks instead

✓ TIPS & BEST PRACTICES

- Use emoji in tab names for quick visual scanning
- Copy a link to a heading (right-click in outline) to send people to the right spot
- Use sub-tabs for appendices so the main tab stays short

QUICK SUMMARY

- `Document outline` — heading navigation
- `Document tabs` — chapters in one file
- `Insert › Table of contents` — refresh manually
- `Section break` — different headers

Writing Faster

@ insert menu · substitutions · Markdown auto-detect · find & replace with regex · dictionary · word count ⌘⇧C · line numbers

TYPE LESS IN DOCS

The @ menu inserts almost anything without leaving the keyboard. **Substitutions** (Tools › Preferences) expand short codes into long text. With **Markdown** turned on, typing # makes a heading and **bold** makes bold. **Find and replace** supports **regular expressions**, which Word's replace doesn't.

NOTES

- Turn on Markdown: Tools › Preferences › **Automatically detect Markdown**
- Substitutions: Tools › Preferences › Substitutions, e.g. `addr` → your address
- Find and replace: tick **Match using regular expressions**; use `$1` in Replace
- Edit › Paste from Markdown / Copy as Markdown converts both ways
- Tools › Line numbers for legal and review docs

SHORTCUTS, MARKDOWN & REGEX

```
@                          # insert menu: chips, tables, dates, blocks
⌘⇧C          / Ctrl+Shift+C    # word count
⌘⌥X          / Ctrl+Alt+X      # spelling & grammar
⌘⇧H          / Ctrl+H          # find & replace

# Markdown auto-detect
# Heading 1    ## Heading 2     - bullet    1. numbered
**bold**   *italic*   `code`   [text](url)   ---

# Find & replace with regex
\s{2,}                    → " "          # collapse spaces
(\w+), (\w+)              → $2 $1        # "Smith, Jane" → "Jane Smith"
^\d+\.\s                  → (empty)      # remove "1. " at line start
```

CODE EXPLAINED

→ Regex in Docs uses RE2 syntax; `$1`, `$2` refer to capture groups

→ Markdown detection works as you type; undo once to keep the literal characters

→ Substitutions are per user and apply in every doc you edit

→ Word count includes a "Display word count while typing" option

OTHER SUITES

⇄ **Microsoft 365** — Word wildcards (not full regex), AutoCorrect entries, Quick Parts

⇄ **Apple iWork** — macOS Text Replacements; Pages Find & Replace without regex

PLATFORM DIFFERENCES

- **Browser** — regex replace, Markdown, substitutions
- **Offline** — same
- **Mobile** — basic find & replace; no regex

REGEX CHEATS

PATTERN	MATCHES
`\d / \w / \s`	Digit / word char / space
`. * + ?`	Any / 0+ / 1+ / optional
`^ $`	Start / end of paragraph
`[A-Z]{2,}`	Two or more capitals
`(…) + $1`	Capture and reuse
`\b`	Word boundary

USEFUL @ ENTRIES

TYPE	INSERTS
`@table`	Table templates
`@checklist`	Checklist
`@code block`	Formatted code block
`@emoji`	Emoji picker
`@timer / @stopwatch`	Timer chips

⚠ COMMON MISTAKES

- Running Replace all with a regex you haven't tested; check matches first
- Leaving Markdown detection on when writing about Markdown
- Creating substitutions that are real words (e.g. `the`)

✓ TIPS & BEST PRACTICES

- Prefix substitution codes with a symbol (`;sig`) to avoid accidents
- Paste from Markdown when moving notes from other apps
- Use line numbers when collecting feedback on a long draft

QUICK SUMMARY

- `@` — insert anything
- `Substitutions` — text expansion
- `Regex replace` — `$1` groups
- `Markdown` — `#`, `**`, `-` as you type

Sheet 09

Review, Mail Merge & Export

Suggesting mode · comment ⌘⌥M · Compare documents · eSignature · mail merge · download .docx / PDF / Markdown · publish to web

FROM DRAFT TO FINAL

Suggesting mode is Docs' track changes: edits appear as suggestions that the owner accepts or rejects. **Compare documents** shows the differences between two docs as suggestions. Workspace plans add **eSignature** and **mail merge** from Sheets. When done, download as Word, PDF, Markdown or EPUB, or **publish to the web**.

NOTES

- Switch mode with the pencil menu (Editing / Suggesting / Viewing) or `⌘⌥⇧X` / `Ctrl+Alt+Shift+X`
- Accept all: Tools › Review suggested edits › Accept all
- Tools › Compare documents picks a second doc and creates a comparison file
- Mail merge (Workspace): Tools › Mail merge, data from a Sheet, send via Gmail
- eSignature (Workspace): Tools › eSignature to add signature fields

AUTOMATE IT

- Mail merge from a Sheet instead of copy-paste letters
- Apps Script merge for free accounts (see sheet 28)

WHERE TO FIND IT

TASK	WHERE / SHORTCUT
Suggesting mode	⌘⌥⇧X / Ctrl+Alt+Shift+X
Editing mode	⌘⌥⇧Z / Ctrl+Alt+Shift+Z
Review suggestions	Tools › Review suggested edits
Compare	Tools › Compare documents
Mail merge	Tools › Mail merge (Workspace)
Download	File › Download
Publish	File › Share › Publish to web

REVIEW A DRAFT

1. Reviewer switches to **Suggesting** (pencil icon, top right)
2. Edits show in colour with the author; comments explain why
3. Owner opens **Tools › Review suggested edits** and steps through
4. Accept or reject each, or Accept all / Reject all
5. Resolve comments as they're handled
6. File › Download › PDF or Word for the final copy

OTHER SUITES

- ⇄ **Microsoft 365** — Word Track Changes, Compare, Mail Merge (desktop)
- ⇄ **Apple iWork** — Pages Track Changes, mail merge, export to Word / PDF / EPUB

PLATFORM DIFFERENCES

- **Browser** — every option
- **Offline** — suggesting works; mail merge and publishing need a connection
- **Mobile** — suggest and comment; no compare or merge

DOWNLOAD FORMATS

FORMAT	KEEPS
.docx	Most formatting, comments, suggestions
PDF	Exact look, links
Markdown (.md)	Headings, lists, links
EPUB	Reflowable e-book
Plain text / .odt / .rtf	Text / open formats

EDITING MODES

MODE	EDITS ARE
Editing	Applied immediately
Suggesting	Proposals with author
Viewing	Not possible (read-only view)
Commenter role	Always suggesting

⚠ COMMON MISTAKES

- Editing in Editing mode on someone else's doc when they wanted suggestions
- Downloading a .docx with unresolved suggestions that the recipient sees as tracked changes
- Publishing to the web and forgetting it stays public until unpublished

✓ TIPS & BEST PRACTICES

- Share as Commenter to force suggestions
- Use Compare documents when edits were made on a copy
- Check File › Share › Publish to web for anything you published earlier

QUICK SUMMARY

- `Suggesting` — Docs track changes
- `Review suggested edits` — accept / reject
- `Compare documents` — diff two docs
- `File › Download` — Word / PDF / Markdown

Sheet 10

Sheets Navigation & Selection

⌘↓ / Ctrl+↓ jump · ⇧ extend · select row / column · Name Box · freeze · switch sheets ⌥↓ / Ctrl+Shift+PgDn · hide / group

MOVE AROUND BIG SHEETS

Sheets uses the same jump logic as Excel: ⌘+Arrow / Ctrl+Arrow goes to the edge of the data block, and adding Shift selects along the way. Type a range or named range in the **Name Box** to jump there. **Freeze** rows and columns so headers stay visible, and **group** rows to collapse detail.

NOTES

- ⌘A / Ctrl+A selects the data region, press again for the whole sheet
- Name Box (left of the formula bar): type D200 or Sales and press Enter
- View › Freeze › 1 row / up to current row; or drag the thick grey bar
- Group rows: select rows › ⌥⇧→ / Alt+Shift+→; collapse with the - button
- Search the menus (⌥/) for "go to range" if the Name Box is hidden

MOVING & SELECTING

```
⌘↑ ↓ ← →        / Ctrl+↑ ↓ ← →      # jump to data edge
⇧⌘↓             / Ctrl+Shift+↓      # select to data edge
fn⌘←            / Ctrl+Home         # go to A1
fn⌘→            / Ctrl+End          # last cell with data
^Space          / Ctrl+Space        # select column
⇧Space          / Shift+Space       # select row
⌘A              / Ctrl+A            # select data / all
⌥↓  ⌥↑          / Ctrl+Shift+PgDn PgUp  # next / previous sheet
⌥⇧K             / Alt+Shift+K       # list of sheets
⌥⇧→  ⌥⇧←        / Alt+Shift+→ ←     # group / ungroup rows or columns
⌘⌥=  ⌘⌥-        / Ctrl+Alt+= -      # insert / delete (rows, columns)
```

CODE EXPLAINED

- → Like Excel, jumps stop at blank cells; gaps in data break them
- → Group collapse / expand buttons appear above columns or left of rows
- → Alt+Shift+K lists all tabs; handy in files with dozens of sheets
- → Turn on compatible spreadsheet shortcuts (in ⌘/) for more Excel-style keys

OTHER SUITES

- ⇄ **Microsoft 365** — Excel uses the same jumps; Go To Special for blanks / visible cells
- ⇄ **Apple iWork** — Numbers tables are objects on a canvas; ⌥↑↓ adds rows

PLATFORM DIFFERENCES

- ▣ **Browser** — every shortcut
- ▣ **Offline** — same
- ▣ **Mobile** — tap / drag only; freeze and group work from menus

FREEZE & GROUP

WANT	DO
Keep header row	View › Freeze › 1 row
Header + first column	Freeze 1 row and 1 column
Up to current cell	View › Freeze › Up to row n
Collapse detail rows	Group (⌥⇧→), click -
Hide rows	⌘⌥9 / Ctrl+Alt+9
Hide columns	⌘⌥0 / Ctrl+Alt+0

WORKING WITH TABS

ACTION	HOW
Rename	Double-click tab
Colour	Right-click › Change colour
Duplicate	Right-click › Duplicate
Copy to another file	Right-click › Copy to
Protect	Right-click › Protect sheet
Hide	Right-click › Hide sheet

⚠ COMMON MISTAKES

- Freezing the wrong rows by selecting a cell lower than intended; use "1 row"
- Gaps in the data so ⌘↓ stops early; fill or delete blank rows
- Hiding rows others need; group them so the collapse is visible

✓ TIPS & BEST PRACTICES

- Colour-code tabs by type (input, calc, output)
- Name key ranges (Data › Named ranges) and jump via the Name Box
- Use ⌥⇧K / Alt+Shift+K instead of scrolling the tab bar

QUICK SUMMARY

- ⌘↓ / Ctrl+↓ — jump to edge
- ⇧ + jump — select
- View › Freeze — sticky headers
- ⌥⇧→ — group rows / columns

Sheet 11

Data Entry, Validation & Cleanup

Smart Fill ⌘⇧Y · fill down ⌘D · ⌘↩ fill selection · dropdowns · checkboxes · split text · remove duplicates · trim whitespace

ENTER FAST, KEEP IT CLEAN

Smart Fill spots patterns (like Flash Fill) and offers to fill the column. **Dropdowns** and **checkboxes** make data entry consistent. The **Data cleanup** menu has **Remove duplicates**, **Trim whitespace** and **Cleanup suggestions**, and **Split text to columns** separates combined values.

NOTES

- Smart Fill: start typing a pattern; accept the grey suggestion with `⌘↩` or use `⌘⇧Y` / `Ctrl+Shift+Y`
- Dropdown: Insert › Dropdown (colour-coded chips) or Data › Data validation
- Checkbox: Insert › Checkbox; stores TRUE / FALSE
- Data › Data cleanup › Cleanup suggestions finds inconsistent values
- Data › Split text to columns guesses the separator; change it if needed

AUTOMATE IT

- Cleanup suggestions handle one-off messes; a recorded macro (sheet 27) repeats the same fix weekly
- Dropdowns from a range (`=Lists!A2:A`) update when the list changes

ENTRY SHORTCUTS

```
⌘D  ⌘R          / Ctrl+D  Ctrl+R      # fill down / right
⌘↩              / Ctrl+Enter          # fill selection with entry
⌘⇧Y             / Ctrl+Shift+Y        # Smart Fill
⌘;              / Ctrl+;              # today's date
⌘⇧;             / Ctrl+Shift+;        # current time
⌘⌥⇧;            / Ctrl+Alt+Shift+;    # date and time
⌘↩ (in cell) / ⌥↩ / Alt+Enter         # new line in a cell
F2 / ↵          / F2 / Enter          # edit cell
⌘K              / Ctrl+K              # insert link

Data validation › Criteria: Dropdown (from a range) → =Lists!$A$2:$A
```

CODE EXPLAINED

- → Fill down copies the top cell into the rest of the selection, adjusting relative references
- → Smart Fill results are formulas you can inspect, not just values
- → An open-ended range (`A2:A`) includes every future row
- → Validation can reject or just warn on invalid input

OTHER SUITES

- ⇄ **Microsoft 365** — Flash Fill `Ctrl+E`, Data Validation lists, Insert › Checkbox
- ⇄ **Apple iWork** — Numbers pop-up menus, checkboxes, autofill handle

PLATFORM DIFFERENCES

- ▣ **Browser** — Smart Fill, cleanup suggestions, chips
- ▣ **Offline** — entry and validation work; suggestions need a connection
- ▣ **Mobile** — dropdowns and checkboxes work; no Smart Fill

CLEANUP TOOLS

TOOL	DOES
Remove duplicates	Deletes repeat rows by chosen columns
Trim whitespace	Removes leading / trailing / double spaces
Cleanup suggestions	Flags inconsistent spellings, types
Column stats	Counts, distribution of one column
Split text to columns	By comma, space, custom

VALIDATION CRITERIA

CRITERIA	EXAMPLE
Dropdown (from a range)	Status list on another tab
Number between	1 and 100
Date is valid / after	after today
Text contains / is email	email check
Custom formula	=COUNTIF(A:A,A2)=1

⚠ COMMON MISTAKES

- Removing duplicates without keeping a copy; it can't be undone after you close the file
- Typing dropdown values into each rule instead of pointing at a list range
- Leaving trailing spaces that break lookups; run Trim whitespace

✓ TIPS & BEST PRACTICES

- Put all dropdown lists on one hidden "Lists" tab
- Use colour-coded dropdown chips for status columns
- Check Column stats before building a pivot

QUICK SUMMARY

- `⌘⇧Y` — Smart Fill
- `⌘↩` — fill selection
- `Insert › Dropdown / Checkbox` — clean entry
- `Data cleanup` — dedupe, trim, suggestions

Formula Basics

HOW SHEETS FORMULAS WORK

Formulas start with = and work like Excel: relative references shift when copied, $ locks them, and F4 cycles the $ while editing. Sheets adds open-ended ranges like `A2:A` (to the bottom) and offers **formula suggestions** and **autocomplete** with argument help. Hover an error to see what went wrong.

NOTES

- Cross-sheet: `=Data!B2`; with spaces: `='Q1 Data'!B2`
- Open-ended ranges: `A2:A` (to the last row), `A2:2` (to the last column)
- Named ranges: Data › Named ranges; use the name in formulas
- Show formulas: View › Show › Formulas, or `^~` / `Ctrl+~`
- Formula help popup: `⇧F1` / `Shift+F1` while typing, or click the ?

REFERENCES & OPERATORS

```
=B2*C2                      # relative: becomes =B3*C3 below
=B2*$F$1                    # locked tax rate
=SUM(Data!B2:B)             # open-ended column on another tab
=SUM('Q1 Data'!B:B)         # tab name with spaces
=Price*(1+TaxRate)          # named ranges
="Total: "&TEXT(D2,"$#,##0")  # → Total: $1,250
=A2<>""                     # TRUE if not blank
=(B2-C2)/C2                 # growth
=IF(ISBLANK(A2), "", A2*2) # blank-safe
```

CODE EXPLAINED

→ `F4` cycles `A1 → $A$1 → A$1 → $A1` while the cursor is on a reference

→ Open-ended ranges are handy, but whole-column math on huge sheets can slow recalculation

→ `&` joins text; wrap numbers in `TEXT()` to control their format

→ Sheets is case-insensitive for function names and text comparisons

OTHER SUITES

⇄ **Microsoft 365** — same syntax; no `A2:A` open ranges (use tables or `A2:.A1000`)

⇄ **Apple iWork** — Numbers uses header names and `Table 1::B2`

PLATFORM DIFFERENCES

- **Browser** — formula suggestions, help popups
- **Offline** — formulas calculate; IMPORT functions wait for a connection
- **Mobile** — edit formulas with a function list

ERROR VALUES

ERROR	USUALLY MEANS
`#N/A`	Lookup found nothing
`#VALUE!`	Wrong type
`#REF!`	Bad / deleted reference, or array would overwrite data
`#DIV/0!`	Divide by zero
`#NAME?`	Unknown function or range name
`#ERROR!`	Formula parse error (bad syntax)
`#NUM!`	Impossible number

OPERATORS

OP	MEANING
`+ - * /`	Arithmetic
`^`	Power
`&`	Join text
`= <> < > <= >=`	Compare
`%`	Percent (=50% → 0.5)
`{1,2;3,4}`	Array literal (sheet 16)

⚠ COMMON MISTAKES

- Forgetting `$` before copying across, so references drift
- Numbers stored as text (left-aligned); use Format › Number or `VALUE()`
- Using `A:A` in a formula placed in column A (circular reference)

✓ TIPS & BEST PRACTICES

- Click a cell while typing a formula instead of typing references by hand
- Name inputs so formulas read as sentences
- Turn off formula suggestions (Tools › Autocomplete) if they get in the way

QUICK SUMMARY

- `$A$1` / `F4` — lock references
- `A2:A` — open-ended range
- `'Tab name'!A1` — cross-sheet
- `^~` — show formulas

Tables & Filter Views

STRUCTURED DATA IN SHEETS

Tables (Format › Convert to table) give a range a name, column **types** (number, date, dropdown, checkbox), banded formatting and **structured references** like `Tasks[Status]`. **Filter views** let each person filter or sort without changing what others see, which matters in shared sheets. **Protected ranges** stop accidental edits.

NOTES

- Convert: select data › `⌘⌥T` / `Ctrl+Alt+T`
- Column types validate entries; dropdown columns show coloured chips
- Group by view: table menu › Create group by view
- Filter (everyone sees it) vs **filter view** (only you, can be saved and shared by link)
- Data › Protect sheets and ranges; choose who can edit

AUTOMATE IT

- Tables + filter views mean one shared sheet works for everyone without copies
- Save filter views like "My open tasks" and bookmark their links

CODE EXPLAINED

- → Structured references keep working when the table grows
- → Filter views have their own URL; share it so people land in that view
- → Group by views summarise a table by one column without a pivot
- → Protecting a range can warn instead of block, for gentle reminders

TABLE REFERENCES & KEYS

```
⌘⌥T            / Ctrl+Alt+T        # convert to table
⌘⇧F            / Ctrl+Shift+F      # hide menus (more room)

=SUM(Tasks[Hours])                   # whole column
=COUNTIF(Tasks[Status], "Done")
=SUMIFS(Tasks[Hours], Tasks[Owner], "Dana")
=XLOOKUP(A2, Products[SKU], Products[Price])
=Tasks[#ALL]  =Tasks[#HEADERS]       # whole table / header row
Data › Create a filter    Data › Filter views › Create new filter view
```

OTHER SUITES

- ⇄ **Microsoft 365** — Excel tables `Ctrl+T`, structured refs, slicers; no per-user filter views (Sheet Views in shared files)
- ⇄ **Apple iWork** — Numbers tables with header rows; filters affect everyone

PLATFORM DIFFERENCES

- ▣ **Browser** — tables, filter views, protection
- ▣ **Offline** — same
- ▣ **Mobile** — use filter views; can't create tables

COLUMN TYPES

TYPE	GIVES
`Number / currency / %`	Format + validation
`Date`	Date picker
`Dropdown`	Coloured chips
`Checkbox`	TRUE / FALSE
`People / smart chip`	Person chips
`Text`	Anything

FILTER VS FILTER VIEW

	FILTER	FILTER VIEW
`Who sees it`	Everyone	Only you
`Saved`	One per sheet	Many, named
`Shareable link`	No	Yes
`Needs edit access`	Yes	No (temporary for viewers)

⚠ COMMON MISTAKES

- Turning on a normal filter in a shared sheet and hiding rows for everyone
- Typing over a table's column type, causing validation warnings
- Protecting the whole sheet and blocking teammates from input cells

✓ TIPS & BEST PRACTICES

- Name tables (click the table name) before writing formulas
- Create filter views for each person or status and link them from a "Start here" tab
- Use warnings, not blocks, for ranges people occasionally need to fix

QUICK SUMMARY

- `⌘⌥T` — convert to table
- `Table[Col]` — structured reference
- `Filter view` — personal filter
- `Protect ranges` — prevent edits

Logic & Math Functions

DECIDE AND SUMMARISE

IF, IFS and SWITCH pick values; the **…IFS** family (SUMIFS, COUNTIFS, AVERAGEIFS, MAXIFS, MINIFS) adds up or counts rows that meet conditions. They behave exactly like Excel's, so formulas move between the two without changes. SUMPRODUCT handles array math and complex conditions.

NOTES

- Criteria are text: `">100"`, `"<>Closed"`, `">="&F1`
- Wildcards in criteria: `"*north*"`; escape with `~`
- SUMIFS: **sum range first**, then range / criteria pairs
- Sheets also has `COUNTUNIQUE` and `COUNTUNIQUEIFS`
- `IFNA` is safer than `IFERROR` around lookups

LOGIC & CONDITIONAL MATH

```
=IF(C2>=70, "Pass", "Fail")
=IFS(C2>=90,"A", C2>=80,"B", C2>=70,"C", TRUE,"F")
=SWITCH(B2, "N","North", "S","South", "Other")
=IF(AND(B2="West", C2>1000), "Big West", "")
=SUMIFS(Sales!D:D, Sales!B:B,"West", Sales!A:A,">="&F1)
=COUNTIFS(A:A,"Open", B:B,"<"&TODAY())    # overdue open items
=AVERAGEIFS(C:C, B:B, "Widget")
=MAXIFS(C:C, B:B, "Widget")               # also MINIFS
=COUNTUNIQUEIFS(Customer, Region, "West")  # distinct customers
=SUMPRODUCT(Qty, Price)                   # Σ qty × price
=ROUND(B2*1.0825, 2)
=IFERROR(B2/C2, 0)
```

CODE EXPLAINED

- → `TRUE, value` at the end of IFS acts as "else"
- → Join cell values into criteria with `&`
- → COUNTUNIQUEIFS counts distinct values that meet conditions (no Excel equivalent)
- → Whole-column refs like `D:D` are fine in Sheets for moderate data sizes

OTHER SUITES

- ⇄ **Microsoft 365** — same functions and order; distinct count via `COUNTA(UNIQUE(FILTER(…)))`
- ⇄ **Apple iWork** — Numbers has IF, IFS, SWITCH, SUMIFS, COUNTIFS, AVERAGEIFS

PLATFORM DIFFERENCES

- ▣ Same everywhere.

CRITERIA EXAMPLES

CRITERIA	MATCHES
`"West"`	Exactly West (any case)
`">100"`	Over 100
`"<>"`	Not blank
`""`	Blank
`"*north*"`	Contains north
`">="&F1`	At least F1

SHEETS EXTRAS

FUNCTION	DOES
`COUNTUNIQUE`	Distinct count
`COUNTUNIQUEIFS`	Distinct count with conditions
`SUMIF with REGEXMATCH`	`=SUMPRODUCT(REGEXMATCH(A2:A,"^N")*B2:B)`
`IFS / SWITCH`	Same as Excel
`MINIFS / MAXIFS`	Same as Excel

⚠ COMMON MISTAKES

- Putting the sum range last (old SUMIF order) in SUMIFS
- Writing `">F1"` instead of `">"&F1`
- IFERROR around everything, hiding broken references

✓ TIPS & BEST PRACTICES

- Drive reports from input cells (`">="&StartDate`)
- Use COUNTUNIQUE for "how many customers" instead of pivots
- Prefer IFS or SWITCH to nested IFs

QUICK SUMMARY

- `IF / IFS / SWITCH` — choose
- `SUMIFS` — sum range first
- `COUNTUNIQUEIFS` — distinct counts
- `IFNA` — lookup misses only

Lookup Functions

XLOOKUP · INDEX / MATCH · VLOOKUP / HLOOKUP · XMATCH · FILTER as a lookup · approximate & wildcard · two-way

PULL VALUES FROM ANOTHER TABLE

Sheets supports **XLOOKUP** with the same arguments as Excel: value, search column, return column, plus optional not-found value and match modes. **INDEX / MATCH** and **VLOOKUP** still work. When you need **every** matching row, use `FILTER` instead of a lookup, and combine with `IMPORTRANGE` to look up in another file.

NOTES

- `XLOOKUP(key, lookup_range, result_range, [missing], [match_mode], [search_mode])`
- match_mode `0` exact, `-1` next smaller, `1` next larger, `2` wildcard
- search_mode `-1` searches from the bottom (latest entry)
- VLOOKUP needs `FALSE` (or `0`) for exact matches
- Cross-file lookups: wrap the other file's range in `IMPORTRANGE` (sheet 19)

LOOKUP PATTERNS

```
=XLOOKUP(A2, Products!A:A, Products!C:C)                # exact
=XLOOKUP(A2, Products!A:A, Products!C:C, "Not found")
=XLOOKUP(A2, Products!A:A, Products!B:D)                # returns 3 columns
=XLOOKUP(D2, Rates!A:A, Rates!B:B, , -1)                # bracket lookup
=XLOOKUP(A2, Log!A:A, Log!C:C, , 0, -1)                 # last match
=INDEX(Products!C:C, MATCH(A2, Products!A:A, 0))        # classic
=VLOOKUP(A2, Products!A:C, 3, FALSE)                    # legacy
=FILTER(Orders!A:D, Orders!B:B = A2)                    # all matches
=XLOOKUP(A2, IMPORTRANGE(url, "P!A:A"), IMPORTRANGE(url, "P!C:C"))
```

CODE EXPLAINED

- → XLOOKUP can return several adjacent columns; they spill right
- → Bracket lookups (`-1`) don't need sorted data in XLOOKUP
- → FILTER returns every matching row, so it replaces "lookup the 2nd match" tricks
- → IMPORTRANGE needs a one-time Allow access click per source file

OTHER SUITES

- ⇄ **Microsoft 365** — identical XLOOKUP / INDEX / MATCH; FILTER also spills
- ⇄ **Apple iWork** — Numbers XLOOKUP, XMATCH, VLOOKUP, INDEX / MATCH

PLATFORM DIFFERENCES

- ▣ Same everywhere; IMPORTRANGE needs a connection.

WHICH LOOKUP?

NEED	USE
One value, exact	XLOOKUP
Value from bracket table	XLOOKUP(…, , -1)
Latest entry	XLOOKUP(…, , 0, -1)
Every matching row	FILTER
Value in another file	XLOOKUP + IMPORTRANGE
Old files	INDEX / MATCH

MATCH MODES

MODE	FINDS
0	Exact (default)
-1	Exact or next smaller
1	Exact or next larger
2	Wildcard * ?
search -1	Bottom-up
search 2 / -2	Binary (sorted)

⚠ COMMON MISTAKES

- VLOOKUP without FALSE returning wrong rows on unsorted data
- Lookup keys with trailing spaces or numbers stored as text
- Separate IMPORTRANGE calls for every row; import once to a tab and look up locally

✓ TIPS & BEST PRACTICES

- Use the missing-value argument instead of IFERROR
- For two conditions, look up a joined key: `XLOOKUP(A2&B2, X:X&Y:Y, Z:Z)`
- Cache IMPORTRANGE data on a hidden tab for speed

QUICK SUMMARY

- `XLOOKUP` — modern lookup
- `, , -1` — next smaller
- `FILTER` — all matches
- `IMPORTRANGE` — other files

Array Formulas

ONE FORMULA FOR A WHOLE COLUMN

Functions like FILTER, SORT, UNIQUE and SEQUENCE return ranges that **spill** into neighbouring cells. ARRAYFORMULA makes ordinary functions work on whole columns, so one formula in the header row replaces thousands of copied formulas. **Array literals** in braces build ranges on the fly: **,** separates columns, **;** separates rows.

NOTES

- `⌘⇧↵` / `Ctrl+Shift+Enter` wraps a formula in ARRAYFORMULA
- Spill blocked by data shows `#REF!` ("Array result was not expanded")
- `{A2:A, C2:C}` puts two columns side by side; `{A2:A; B2:B}` stacks them
- In some locales the column separator is `\` instead of `,`
- `SORTN` returns the top N rows, with tie handling

ARRAY FORMULAS

```
=ARRAYFORMULA(IF(A2:A="", "", B2:B*C2:C))      # whole-column total
=ARRAYFORMULA(PROPER(TRIM(A2:A)))               # clean a column
=FILTER(A2:D, C2:C="West", D2:D>1000)           # multiple conditions = AND
=FILTER(A2:D, (C2:C="West")+(C2:C="East"))      # OR
=SORT(A2:D, 4, FALSE)                           # by 4th column, descending
=SORTN(A2:D, 5, 0, 4, FALSE)                    # top 5 by column 4
=UNIQUE(B2:B)          =COUNTA(UNIQUE(B2:B))
=SEQUENCE(12)          =SEQUENCE(1, 7, 0)
={A1:A10, C1:C10}                               # two columns together
=VSTACK(Jan!A2:D, Feb!A2:D)                     # stack tabs
=CHOOSECOLS(A2:F, 1, 4)                         # pick columns
=TOCOL(A2:F, 1)                                 # flatten, skip blanks
```

CODE EXPLAINED

- → The `IF(A2:A="", "", …)` guard stops ARRAYFORMULA filling empty rows with zeros
- → FILTER takes extra condition arguments for AND; add conditions for OR
- → SORT's order argument is TRUE / FALSE (ascending / descending)
- → `TOCOL(range, 1)` ignores blanks; `FLATTEN` is the older equivalent

OTHER SUITES

- ⇄ **Microsoft 365** — dynamic arrays spill natively; no ARRAYFORMULA needed
- ⇄ **Apple iWork** — recent Numbers adds FILTER, SORT, SORTBY, UNIQUE, SEQUENCE; no ARRAYFORMULA

PLATFORM DIFFERENCES

▣ Same everywhere.

ARRAY FUNCTIONS

FUNCTION	DOES
`ARRAYFORMULA(expr)`	Apply to whole ranges
`FILTER(range, cond1, …)`	Matching rows
`SORT(range, col, asc)`	Sort
`SORTN(range, n, …)`	Top n
`UNIQUE(range)`	Distinct rows
`SEQUENCE(rows, cols, start, step)`	Number grid
`VSTACK / HSTACK`	Combine
`CHOOSECOLS / CHOOSEROWS`	Pick
`TOCOL / TOROW / WRAPROWS`	Reshape

ARRAY LITERALS

LITERAL	GIVES
`{1, 2, 3}`	One row, three columns
`{1; 2; 3}`	One column, three rows
`{A:A, C:C}`	Columns side by side
`{"Name", "Total"; A2:A, D2:D}`	Header row on top (must match widths)
`{A2:A; B2:B}`	Stack two columns

⚠ COMMON MISTAKES

- Copying an ARRAYFORMULA down the column; it already fills it
- Typing in a cell inside a spill range, causing `#REF!` on the formula
- Array literal pieces of different heights or widths (`#VALUE!` / mismatched)

✓ TIPS & BEST PRACTICES

- Put ARRAYFORMULA in the header row (`{"Total"; ARRAYFORMULA(…)}`) so sorting never breaks it
- Use FILTER + SORTN for leaderboards
- Prefer VSTACK over array literals for readability

QUICK SUMMARY

- `ARRAYFORMULA` — whole-column formulas
- `FILTER / SORT / UNIQUE` — spill results
- `{a, b}` / `{a; b}` — columns / rows
- `SORTN` — top N

Sheet 17

The QUERY Function

QUERY(range, "select … where … group by …") · Col1 vs A · order by · pivot · label / format · headers · dates

SQL INSIDE A CELL

QUERY runs a SQL-like statement (Google Visualization Query Language) over a range: **select** columns, filter with **where**, summarise with **group by**, sort with **order by**, cross-tab with **pivot**. One QUERY often replaces a pivot table, a FILTER and several SUMIFS.

NOTES

- Column names are letters (`A` , `B`) for ranges; `Col1` , `Col2` for arrays and IMPORTRANGE
- Text values in single quotes: `where B = 'West'`
- Clause order matters: select, where, group by, pivot, order by, limit, offset, label, format
- Third argument = number of header rows (usually `1`)
- Dates need the `date` keyword: `where A >= date '2026-01-01'`

AUTOMATE IT

- One QUERY on a raw-data tab instead of a chain of helper columns and pivots
- Build the date in the where clause from a cell so reports update by changing one input

QUERY EXAMPLES

```
=QUERY(Sales!A:E, "select A, B, E where B = 'West' order by E desc", 1)
=QUERY(Sales!A:E, "select B, sum(E) group by B order by sum(E) desc", 1)
=QUERY(Sales!A:E, "select B, sum(E) group by B pivot C", 1)      # cross-tab
=QUERY(Sales!A:E, "select * where E > 1000 limit 10", 1)
=QUERY(Sales!A:E, "select B, count(A) group by B label count(A) 'Orders'", 1)
=QUERY(Sales!A:E, "select A, E where A >= date '"&TEXT(G1,"yyyy-mm-dd")&"'", 1)
=QUERY(Sales!A:E, "select B where D contains 'widget'", 1)
=QUERY(IMPORTRANGE(url, "Data!A:E"), "select Col2, sum(Col5) group by Col2", 1)
=QUERY({Jan!A2:E; Feb!A2:E}, "select Col1, Col5 where Col5 is not null", 0)
```

CODE EXPLAINED

- → Aggregates: `sum` , `avg` , `count` , `max` , `min` ; every non-aggregated selected column must be in group by
- → Joining a cell into the query string needs quote juggling: `'"&G1&"'`
- → Arrays (from `{}` or IMPORTRANGE) use `Col1` , not `A`
- → `contains` , `starts with` , `matches` (regex) and `like` work on text

OTHER SUITES

- ⇄ **Microsoft 365** — no QUERY; use FILTER + GROUPBY / PIVOTBY or Power Query
- ⇄ **Apple iWork** — no equivalent; use pivot tables and categories

PLATFORM DIFFERENCES

- ▣ Same everywhere; QUERY over IMPORTRANGE needs a connection.

CLAUSES (IN ORDER)

CLAUSE	EXAMPLE
`select`	`select A, sum(E)`
`where`	`where B = 'West' and E > 0`
`group by`	`group by A`
`pivot`	`pivot C`
`order by`	`order by sum(E) desc`
`limit / offset`	`limit 10 offset 5`
`label`	`label sum(E) 'Total'`
`format`	`format sum(E) '$#,##0'`

WHERE OPERATORS

OPERATOR	EXAMPLE
`= != < > <= >=`	`E >= 100`
`and / or / not`	`B = 'W' or B = 'E'`
`contains`	`D contains 'pro'`
`starts with / ends with`	`A starts with 'INV'`
`matches`	`A matches '^\d{4}$'`
`is null / is not null`	`E is not null`

⚠ COMMON MISTAKES

- Double quotes around text inside the query; use single quotes
- Selecting a column without aggregating or grouping it
- Mixed data types in one column; QUERY uses the majority type and blanks the rest

✓ TIPS & BEST PRACTICES

- Build long queries in a cell and reference it: `=QUERY(A:E, H1, 1)`
- Use `label` to give result columns friendly headers
- See the SQL set for the full query mindset

QUICK SUMMARY

- `select … where … group by` — SQL in a cell
- `'text'` — single quotes
- `Col1` — for arrays / IMPORTRANGE
- `pivot` — cross-tab

Sheet 18

Text, Date & Regex Functions

SPLIT · JOIN / TEXTJOIN · TEXT · REGEXMATCH / REGEXEXTRACT / REGEXREPLACE · TODAY · EOMONTH · NETWORKDAYS · DATEDIF

CLEAN TEXT, DO DATE MATH

Sheets has had regular expressions for years: REGEXMATCH, REGEXEXTRACT and REGEXREPLACE use the RE2 syntax. SPLIT breaks text into columns and TEXTJOIN joins ranges. Dates are serial numbers like in Excel, so subtraction gives days and EOMONTH, EDATE, NETWORKDAYS handle calendars.

NOTES

- REGEX functions are case-sensitive; add `(?i)` at the start of the pattern to ignore case
- `SPLIT(text, delimiter)` splits on each character of the delimiter unless `split_by_each` is FALSE
- `TEXT(value, "format")` returns formatted text
- Dates: Format › Number › Date; `=B2-A2` gives days
- `DATEDIF` works and is documented in Sheets

TEXT & DATE FORMULAS

```
=SPLIT("Smith, Jane", ", ", FALSE)          # Smith | Jane
=TEXTJOIN(", ", TRUE, B2:B10)               # join, skip blanks
=PROPER(TRIM(A2))
=REGEXMATCH(A2, "^\d{5}$")                   # TRUE for 5-digit zip
=REGEXEXTRACT(A2, "[\w.]+@[\w.]+")           # pull an email
=REGEXEXTRACT(A2, "(\d{3})-(\d{4})")         # two groups → two columns
=REGEXREPLACE(A2, "(?i)inc\.?", "Inc.")      # case-insensitive replace
=TEXT(B2, "$#,##0.00")     =TEXT(C2, "mmm d, yyyy")
=EOMONTH(TODAY(), 0)                        # month end
=EDATE(A2, 3)                               # +3 months
=NETWORKDAYS(A2, B2, Holidays!A:A)
=DATEDIF(Birth, TODAY(), "Y")               # age
=WORKDAY(TODAY(), 10)
```

CODE EXPLAINED

- → REGEXEXTRACT with several capture groups spills one column per group
- → In Sheets formulas a single backslash is enough: `"\d"`
- → `SPLIT(…, ", ", FALSE)` splits on the whole ", " string
- → `EOMONTH(d, -1)+1` = first day of the month

OTHER SUITES

- ⇄ **Microsoft 365** — REGEXTEST / REGEXEXTRACT / REGEXREPLACE in M365, TEXTSPLIT, TEXTJOIN
- ⇄ **Apple iWork** — Numbers REGEX inside functions, REGEX.EXTRACT, TEXTJOIN, DATEDIF

PLATFORM DIFFERENCES

- ▣ Same everywhere.

RE2 CHEATS

PATTERN	MATCHES
`\d \w \s`	Digit / word / space
`[A-Z]{2,}`	2+ capitals
`^ $`	Start / end
`(…)`	Capture group
`(?i)`	Ignore case
`.*?`	Shortest match
`no lookbehind`	RE2 doesn't support it

DATE FUNCTIONS

FUNCTION	RETURNS
`TODAY() / NOW()`	Date / date + time
`DATE(y, m, d)`	Build a date
`WEEKDAY(d, 2)`	1 = Monday
`WEEKNUM / ISOWEEKNUM`	Week number
`EOMONTH / EDATE`	Month math
`NETWORKDAYS / WORKDAY`	Business days
`DATEDIF(a, b, "M")`	Whole months

⚠ COMMON MISTAKES

- Dates typed in a format your locale doesn't read become text; check File › Settings › Locale
- Expecting REGEXMATCH to ignore case; add `(?i)`
- Using lookbehind `(?<=…)`, which RE2 doesn't support

✓ TIPS & BEST PRACTICES

- Use REGEXEXTRACT instead of MID / FIND chains
- Keep a Holidays tab for NETWORKDAYS
- Wrap regex functions in ARRAYFORMULA to process a whole column

QUICK SUMMARY

- `SPLIT / TEXTJOIN` — split / join
- `REGEXMATCH / EXTRACT / REPLACE` — RE2 patterns
- `(?i)` — ignore case
- `EOMONTH / NETWORKDAYS` — date math

Import & Live Data

PULL DATA FROM OTHER FILES AND THE WEB

Sheets can fetch live data. `IMPORTRANGE` pulls a range from another spreadsheet you can access. `IMPORTHTML` grabs tables and lists from web pages, `IMPORTXML` uses XPath, and `IMPORTDATA` loads CSV files. `GOOGLEFINANCE` returns stock prices and history, `GOOGLETRANSLATE` translates text, and `IMAGE` puts a picture in a cell.

NOTES

- IMPORTRANGE asks once per source file: click **Allow access**
- Import functions refresh on a schedule (roughly hourly) and when the sheet opens
- File › Settings › Calculation controls recalculation for NOW / RAND / GOOGLEFINANCE
- Pages that build content with JavaScript can't be read by IMPORTHTML / IMPORTXML
- Insert › Image › Image in cell, or `=IMAGE(url)`

AUTOMATE IT

- Use IMPORTRANGE to keep one summary file in sync with each team's sheet
- Import once to a hidden tab, then reference that tab, to limit fetches

IMPORT FUNCTIONS

```
=IMPORTRANGE("https://docs.google.com/spreadsheets/d/ID", "Data!A1:E500")
=IMPORTRANGE(B1, "Data!A:E")                    # URL or ID in a cell
=IMPORTHTML("https://example.com/stats", "table", 2)   # 2nd table on page
=IMPORTXML("https://example.com", "//h2")       # all h2 headings (XPath)
=IMPORTDATA("https://example.com/data.csv")
=GOOGLEFINANCE("NASDAQ:GOOG")                   # current price
=GOOGLEFINANCE("MSFT", "close", DATE(2026,1,1), TODAY(), "WEEKLY")
=GOOGLEFINANCE("CURRENCY:USDEUR")               # exchange rate
=GOOGLETRANSLATE(A2, "en", "es")
=IMAGE("https://example.com/logo.png", 1)       # fit to cell
=DETECTLANGUAGE(A2)
```

CODE EXPLAINED

- → IMPORTRANGE accepts the full URL or just the spreadsheet ID
- → IMPORTHTML's index counts tables (or lists) from 1
- → GOOGLEFINANCE history returns a Date / Close table that spills
- → IMAGE modes: 1 fit, 2 stretch, 3 original size, 4 custom size

OTHER SUITES

- ⇄ **Microsoft 365** — Power Query (Get Data › From Web / Workbook), STOCKHISTORY, linked data types
- ⇄ **Apple iWork** — no web imports in Numbers; `STOCK` / `STOCKH` for market data

PLATFORM DIFFERENCES

- ▣ **Browser** — all functions
- ▣ **Offline** — last fetched values show; no refresh
- ▣ **Mobile** — values display and refresh online

IMPORT FUNCTIONS

FUNCTION	SOURCE
`IMPORTRANGE`	Another Google Sheet
`IMPORTHTML`	HTML table / list
`IMPORTXML`	Any XPath on a page, XML / RSS
`IMPORTDATA`	CSV / TSV URL
`IMPORTFEED`	RSS / Atom feed
`GOOGLEFINANCE`	Stocks, funds, currency
`IMAGE`	Picture from URL

XPATH STARTERS

XPATH	GETS
`//h1`	All h1 text
`//a/@href`	All link URLs
`//table[1]//tr`	Rows of the first table
`//*[@class='price']`	Elements with class price
`//meta[@name='description']/@content`	Meta description

⚠ COMMON MISTAKES

- Hundreds of IMPORTRANGE calls slowing the file; import once and reference locally
- Expecting IMPORTXML to read sites that render with JavaScript
- Using GOOGLEFINANCE for trading decisions; data can be delayed up to 20 minutes

✓ TIPS & BEST PRACTICES

- Keep source URLs in cells so swaps are one edit
- Wrap imports in IFERROR to show a friendly message while loading
- Combine IMPORTRANGE with QUERY to fetch only the rows you need

QUICK SUMMARY

- `IMPORTRANGE` — other sheets
- `IMPORTHTML / IMPORTXML` — web pages
- `GOOGLEFINANCE` — market data
- `IMAGE` — picture in a cell

LET, LAMBDA & Named Functions

YOUR OWN FUNCTIONS, NO SCRIPT NEEDED

LET names steps inside a formula. LAMBDA defines a function, and the helper functions MAP, BYROW, BYCOL, REDUCE, SCAN and MAKEARRAY apply it across ranges. Save a LAMBDA as a **Named function** (Data › Named functions) to call it like `=TAXED(B2)` anywhere in the file, and import named functions from other files.

NOTES

- Named functions: Data › Named functions › Add new function
- Give each argument a placeholder name and description; they show in formula help
- Import from another file: Named functions › Import function
- BYROW / BYCOL return one value per row / column
- Named functions can call other named functions

LET, LAMBDA & HELPERS

```
=LET(total, SUM(D2:D), west, SUMIFS(D2:D, B2:B, "West"), west/total)

# Data › Named functions → name: TAXED, args: amount, rate
#   definition: =ROUND(amount*(1+rate), 2)
=TAXED(B2, 0.0825)                              # 100 → 108.25

=MAP(A2:A20, LAMBDA(x, PROPER(TRIM(x))))        # clean each cell
=BYROW(B2:E20, LAMBDA(r, MAX(r)-MIN(r)))        # range per row
=BYCOL(B2:E20, LAMBDA(c, AVERAGE(c)))           # average per column
=REDUCE(0, C2:C20, LAMBDA(acc, x, acc + (x>100)))  # count over 100
=SCAN(0, C2:C20, LAMBDA(acc, x, acc + x))       # running total
=MAKEARRAY(5, 5, LAMBDA(r, c, r*c))             # times table
=LAMBDA(x, x*2)(21)                             # → 42
```

CODE EXPLAINED

- → Named functions don't include the `LAMBDA(` wrapper; arguments are declared in the side panel
- → MAP passes each cell; BYROW passes a whole row as a range
- → SCAN is REDUCE that returns every intermediate value
- → Unlike Apps Script custom functions, named functions recalculate instantly

OTHER SUITES

- ⇄ **Microsoft 365** — LET, LAMBDA, named LAMBDAs in Name Manager, same helpers
- ⇄ **Apple iWork** — recent Numbers has LET, LAMBDA and SCAN; no named functions

PLATFORM DIFFERENCES

- ▣ Same everywhere.

LAMBDA HELPERS

FUNCTION	RETURNS
`MAP(arr, λ)`	Same-size array
`BYROW(arr, λ)`	One value per row
`BYCOL(arr, λ)`	One value per column
`REDUCE(init, arr, λ)`	Single accumulated value
`SCAN(init, arr, λ)`	Running values
`MAKEARRAY(r, c, λ)`	Generated grid

NAMED FUNCTION VS APPS SCRIPT

	NAMED FN	APPS SCRIPT FN
`Language`	Formulas	JavaScript
`Speed`	Instant	Slower (server call)
`Can call APIs`	No	Yes
`Shareable`	Import between files	Copy the script

⚠ COMMON MISTAKES

- Wrapping a named function's definition in another LAMBDA
- Argument names that clash with cell references (`A1`, `B2`)
- Using Apps Script for something a named function does faster

✓ TIPS & BEST PRACTICES

- Add descriptions and examples to named functions for teammates
- Refactor long formulas into LET first, then into a named function
- Keep a "function library" sheet and import from it

QUICK SUMMARY

- `LET` — name steps
- `LAMBDA` — define a function
- `Named functions` — reusable in the file
- `MAP / BYROW / REDUCE / SCAN` — loop over ranges

Sheet 21

Pivot Tables, Charts & Conditional Formatting

pivot tables · suggested pivots · calculated fields · chart editor · SPARKLINE · custom formula rules · color scales · Connected Sheets

SUMMARISE AND HIGHLIGHT

Pivot tables (Insert › Pivot table) summarise data by dragging fields into Rows, Columns, Values and Filters, and Sheets suggests pivots for you. The **chart editor** builds charts that update with the data, and SPARKLINE draws a tiny chart in a cell. **Conditional formatting** with a **custom formula** highlights whole rows by any rule.

NOTES

- Pivot source: use an open-ended range (`A1:E`) or a table so new rows are included
- Calculated field: Values › Add › Calculated field, e.g. `=Revenue-Cost`
- Group dates: right-click a date in the pivot › Create pivot date group
- Custom formula rules use the top-left cell of the range with relative refs
- **Connected Sheets** (Workspace) analyses BigQuery data with pivots

AUTOMATE IT

- Pivot on an open-ended range so new rows show up automatically
- A time-driven Apps Script trigger can email a chart or PDF weekly

RULES, SPARKLINES & CHARTS

```
# Format › Conditional formatting › Custom formula is
=$E2="Overdue"                          # whole row when E is Overdue
=AND($D2<TODAY(), $F2<>"Done")          # past due, not done
=COUNTIF($A:$A, $A2)>1                  # duplicates
=ISEVEN(ROW())                          # banded rows

# SPARKLINE in a cell
=SPARKLINE(B2:M2)                                          # line
=SPARKLINE(B2:M2, {"charttype","column"; "color","#4285f4"})
=SPARKLINE(C2, {"charttype","bar"; "max", 100})            # progress bar
=SPARKLINE(B2:M2, {"charttype","winloss"})

Insert › Chart · Insert › Pivot table · Explore (✦) suggests both
```

CODE EXPLAINED

- → `$E2` locks the column so every cell in the row checks column E
- → SPARKLINE options go in a 2-column array literal of name / value pairs
- → A bar sparkline with `max` makes a simple progress bar
- → Charts and pivots update as the data changes; no refresh needed

OTHER SUITES

- ⇄ **Microsoft 365** — PivotTables (refresh needed), sparklines via Insert, formula-based CF
- ⇄ **Apple iWork** — Numbers pivot tables, interactive charts, conditional highlighting (no formulas)

PLATFORM DIFFERENCES

- ▣ **Browser** — everything; Connected Sheets on Workspace
- ▣ **Offline** — pivots and charts work
- ▣ **Mobile** — view and light edits; create pivots on tablets

CONDITIONAL FORMAT TYPES

RULE	EXAMPLE
`Single colour: value`	Greater than 100
`Text contains`	"urgent"
`Date is`	Before today
`Custom formula`	Any TRUE / FALSE formula
`Colour scale`	Heat map, min / mid / max

SPARKLINE TYPES

CHARTTYPE	SHOWS
`line (default)`	Trend
`column`	Bars per value
`bar`	One stacked / progress bar
`winloss`	Positive vs negative
`Options`	color, max, min, linewidth

⚠ COMMON MISTAKES

- Custom formula written for row 1 when the range starts at row 2
- Pivots on a fixed range (`A1:E500`) that miss new rows
- Too many conditional rules on large sheets, making them slow

✓ TIPS & BEST PRACTICES

- Try **Explore** (bottom right) for instant pivot and chart suggestions
- Use SPARKLINE progress bars in task trackers
- Name pivot tabs clearly; they're easy to lose among data tabs

QUICK SUMMARY

- `Insert › Pivot table` — summaries
- `=$E2="x"` — row-based rule
- `SPARKLINE` — chart in a cell
- `Explore` — suggestions

Sheet 22

Themes, Master & Building Slides

Edit theme · import theme · Import slides · duplicate ⌘D · align & distribute · group ⌘⌥G · linked charts · animations

CONSISTENT DECKS WITHOUT THE FUSS

Every Slides deck has a **theme** with a master slide and layouts. **Slide › Edit theme** opens the theme builder: set fonts, colours and logo once and every slide follows. **Import slides** pulls slides from another deck, and **linked charts** from Sheets update with one click. The **Motion** pane holds transitions and animations.

NOTES

- Edit theme: Slide › Edit theme (or View › Theme builder)
- Import theme: Slide › Change theme › Import theme
- Import slides: File › Import slides, pick a deck and slides
- Linked chart: copy a chart in Sheets, paste in Slides › Link to spreadsheet
- Animations: select object › Insert › Animation (Motion pane)

AUTOMATE IT

- Master slides and Import slides instead of rebuilding standard slides each time
- Linked charts update from Sheets with one click (Update button)

BUILDING SHORTCUTS

```
⌘M              / Ctrl+M              # new slide
⌘D              / Ctrl+D              # duplicate slide or object
⌘⌥G             / Ctrl+Alt+G          # group
⌘⌥⇧G            / Ctrl+Alt+Shift+G    # ungroup
⌘↑ / ⌘↓         / Ctrl+↑ / Ctrl+↓     # bring forward / send backward
⌘⇧↑ / ⌘⇧↓       / Ctrl+Shift+↑ / ↓    # bring to front / send to back
⇧ + arrow       / Shift + arrow       # nudge by one pixel
⌥ + drag        / Alt + drag          # duplicate while dragging
⌘⌥⇧B            / Ctrl+Alt+Shift+B    # Motion (animations) pane
```

CODE EXPLAINED

- → `⌘D` on an object duplicates it in place; move it, then `⌘D` again to repeat the offset
- → Arrange › Align and Arrange › Distribute work on two or more objects
- → Linked charts show an Update button when the Sheet changes
- → Hold `⇧` while resizing to keep proportions

OTHER SUITES

- ⇄ **Microsoft 365** — Slide Master, Reuse Slides, Morph, Designer
- ⇄ **Apple iWork** — Keynote master slides, Magic Move, themes

PLATFORM DIFFERENCES

- ▣ **Browser** — theme builder, import, animations
- ▣ **Offline** — editing works; linked chart updates need a connection
- ▣ **Mobile** — edit text and layouts; no theme builder

THEME BUILDER

EDIT	AFFECTS
`Theme (top slide)`	All layouts
`A layout`	Slides using it
`Colours`	Theme palette in every menu
`Fonts`	Placeholders
`Rename layout`	Easier picking in Layout menu

ARRANGE MENU

COMMAND	USE
`Align left / center / right`	Line up edges
`Distribute horizontally`	Equal gaps
`Center on page`	Horizontal / vertical
`Order`	Front / back
`Rotate / flip`	Turn objects

⚠ COMMON MISTAKES

- Formatting text on each slide instead of in the theme builder
- Pasting charts as images; they go stale when the data changes
- Leaving unused layouts that clutter the Layout menu

✓ TIPS & BEST PRACTICES

- Import slides from a "library" deck of approved slides
- Turn on View › Guides for consistent margins
- Use the theme's colour palette so recolouring later is one change

QUICK SUMMARY

- `Slide › Edit theme` — master + layouts
- `File › Import slides` — reuse slides
- `⌘D` / `Ctrl+D` — duplicate
- `Linked chart` — Update from Sheets

Presenting

PRESENT FROM THE BROWSER

Slideshow (⌘↩ / `Ctrl+Enter`) goes full screen; **Presenter view** opens a second window with notes, a timer and next-slide preview. **Audience Q&A** lets people submit questions from a link shown on the slide. Turn on **captions** to show live subtitles. Present straight into a **Meet** call or publish the deck to the web.

NOTES

- Presenter view: Slideshow dropdown › Presenter view
- Audience Q&A: Presenter view › Audience tools › Start new
- Speaker notes: View › Show speaker notes; voice-type into them
- Present in Meet from the Present button inside the call
- File › Download › PDF or .pptx; File › Share › Publish to web for auto-advancing embeds

PRESENTATION KEYS

```
⌘↩              / Ctrl+Enter          # present from current slide
⌘⇧↩             / Ctrl+Shift+Enter    # present from the beginning
→ ↓ Space       / → ↓ Space           # next
← ↑             / ← ↑                 # previous
12 ↩            / 12 Enter            # go to slide 12
B   W           / B   W               # black / white screen
L               / L                   # laser pointer
S               / S                   # open speaker notes
A               / A                   # audience tools
⌘⇧C             / Ctrl+Shift+C        # captions on / off
Esc             / Esc                 # exit
```

CODE EXPLAINED

- → Presenter view opens in a new browser window; drag it to your laptop screen
- → The Q&A link appears at the top of every slide while active
- → Captions use your microphone and work in several languages
- → Published decks can auto-advance and loop for kiosks

OTHER SUITES

- ⇄ **Microsoft 365** — PowerPoint Presenter View, Speaker Coach, live subtitles
- ⇄ **Apple iWork** — Keynote presenter display, `X` swap displays, iPhone remote

PLATFORM DIFFERENCES

- ▣ **Browser** — Presenter view, Q&A, captions
- ▣ **Offline** — present marked-offline decks; no Q&A or captions
- ▣ **Mobile** — present to a TV / cast; phone as remote via Meet

SHARE & EXPORT

OPTION	WHERE
PDF	File › Download › PDF
PowerPoint	File › Download › .pptx
Image per slide	File › Download › PNG / JPEG
Publish (auto-play)	File › Share › Publish to web
Present in Meet	Meet › Present › a tab / Slides

BEFORE YOU PRESENT

CHECK	WHY
Make available offline	Bad venue Wi-Fi
Download PDF backup	If the browser fails
Close other tabs	Fewer notifications
Test Presenter view	Right screen for notes
Embedded videos load	Cached and permissions OK

⚠ COMMON MISTAKES

- Presenting the editor window instead of the slideshow
- Forgetting to end Audience Q&A, leaving the banner on later slides
- Relying on Wi-Fi for embedded videos

✓ TIPS & BEST PRACTICES

- Type a slide number and Enter to jump during Q&A
- Press `B` to blank the screen for discussion
- Keep a PDF copy on your laptop as a fallback

QUICK SUMMARY

- `⌘↩` / `Ctrl+Enter` — present
- `Presenter view` — notes + timer
- `L / B / W` — laser, blank screens
- `⌘⇧C` — captions

Gmail Efficiency

keyboard shortcuts on · j / k / e / r / a / c · search operators · labels & filters · templates · snooze · schedule send · multiple inboxes

INBOX ZERO, THE GMAIL WAY

Turn on **keyboard shortcuts** (Settings › See all settings › Keyboard shortcuts on) and Gmail becomes keyboard-driven: `j` / `k` to move, `e` to archive, `r` to reply. **Search operators** find anything, and any search can become a **filter** that labels, archives or forwards future mail automatically. **Templates** store canned replies.

NOTES

- `?` shows every shortcut once they're enabled
- Labels are like folders, but a message can have several
- Create a filter: search, then the sliders icon › Create filter
- Templates: Settings › Advanced › Templates on; then compose › ⋯ › Templates
- Undo send: Settings › General › up to 30 seconds

AUTOMATE IT

- Filters that label, skip the inbox or forward mail matching a search
- Templates for replies you send more than twice a week

GMAIL KEYS & SEARCH

```
c           # compose              r / a / f    # reply / reply all / forward
j / k       # next / previous      o or ↵       # open
e           # archive              #            # delete
s           # star                 l / v        # label / move to
x           # select               ⇧I / ⇧U      # mark read / unread
b           # snooze               /            # search
g i         # go to inbox          g s          # go to starred
⌘↵ / Ctrl+Enter                    # send

from:dana@acme.com has:attachment larger:5M older_than:1y
subject:(invoice OR receipt) after:2026/09/01 -label:paid
is:unread in:inbox category:primary
```

CODE EXPLAINED

- → Single-key shortcuts only work after you turn them on in Settings
- → `g` then a letter jumps to a view (`g i` inbox, `g t` sent)
- → Dates in search use `YYYY/MM/DD`; `older_than:1y` / `newer_than:7d` are relative
- → `-` excludes, `OR` (capitals) and parentheses group

OTHER SUITES

- ⇄ **Microsoft 365** — Outlook search operators, Quick Steps, rules, My Templates
- ⇄ **Apple iWork** — Apple Mail Smart Mailboxes, rules, Remind Me, Send Later

PLATFORM DIFFERENCES

- ▣ **Browser** — every shortcut and filter
- ▣ **Offline** — Gmail offline mode (Chrome) reads and queues mail
- ▣ **Mobile** — swipe actions, snooze, schedule send; no shortcuts or filters editor

SEARCH OPERATORS

OPERATOR	FINDS
`from: / to: / cc:`	People
`subject:`	Words in subject
`has:attachment / filename:pdf`	Attachments
`larger:10M`	Big messages
`older_than:1y / newer_than:7d`	Relative dates
`label: / -label:`	With / without label
`is:unread / is:starred`	State

INBOX TOOLS

TOOL	USE
`Snooze (b)`	Come back later
`Schedule send`	Send dropdown › Schedule
`Multiple inboxes`	Sections by search
`Categories`	Primary / Promotions / Updates
`Nudges`	Reminders to reply
`Confidential mode`	Expiry, no forwarding

⚠ COMMON MISTAKES

- Pressing letter keys while shortcuts are on and archiving mail by accident (undo with `z`)
- Using labels like folders and never archiving, so the inbox stays full
- Filters that skip the inbox without a label, so mail vanishes

✓ TIPS & BEST PRACTICES

- Use `e` (archive) as your default action; search finds everything later
- Turn any useful search into a filter
- Set undo send to 30 seconds

QUICK SUMMARY

- `j k e r` — move, archive, reply
- `from: has:attachment older_than:` — search
- `Filters` — automatic labels / archive
- `Templates` — canned replies

Calendar & Meet

SCHEDULE AND MEET FASTER

Google Calendar has single-key shortcuts (`c` create, `t` today, `d` / `w` / `m` views). **Find a time** shows attendees' availability side by side; **appointment schedules** give people a booking page. **Working hours & location** tells others when and where you work. In **Meet**, shortcuts handle mute, camera and hand raise.

NOTES

- Shortcuts are on by default in Calendar; `?` lists them
- Appointment schedules: Create › Appointment schedule (booking page; more options on paid plans)
- Settings › Working hours & location; add focus time and out of office
- **Time insights** (left panel) shows how your time is split
- Meet: Activities › Recording / Transcripts / Take notes for me (plan dependent)

AUTOMATE IT

- Appointment schedule link in your email signature instead of back-and-forth
- Focus time events that auto-decline meetings

CALENDAR & MEET KEYS

```
# Calendar
c              # create event          e      # event details
t              # today                /      # search
d w m y        # day, week, month, year   x   # custom view
j / k  or n / p   # next / previous period
q              # quick add             ⌫ / Delete    # delete event

# Meet
⌘D             / Ctrl+D          # mute / unmute
⌘E             / Ctrl+E          # camera on / off
^⌘H            / Ctrl+Alt+H      # raise / lower hand
CC button      / CC button       # captions
```

CODE EXPLAINED

- → Quick add (`q` or typing in the event box) understands "Lunch with Sam Friday 12pm"
- → `x` cycles your custom view (e.g. 4 days)
- → Meet shortcuts work while the Meet tab has focus
- → Out of office events auto-decline meetings during that time

OTHER SUITES

- ⇄ **Microsoft 365** — Outlook Scheduling Assistant, Bookings, Teams shortcuts
- ⇄ **Apple iWork** — Apple Calendar quick event, FaceTime links

PLATFORM DIFFERENCES

- ▣ **Browser** — every feature
- ▣ **Offline** — view events when offline mode is on
- ▣ **Mobile** — Calendar and Meet apps; booking pages work in any browser

SCHEDULING TOOLS

TOOL	USE
`Find a time`	Free / busy side by side
`Suggested times`	Auto-picked slots
`Appointment schedule`	Booking page
`Out of office`	Auto-decline
`Focus time`	Block + auto-decline
`Working location`	Office / home / other

MEET FEATURES

FEATURE	WHERE
`Background blur`	Visual effects
`Captions / translation`	CC button
`Recording`	Activities (paid plans)
`Take notes for me`	Gemini (plan dependent)
`Breakout rooms / polls`	Activities

⚠ COMMON MISTAKES

- Booking across time zones without checking the attendee's zone
- Leaving appointment schedules open with no buffer between meetings
- Forgetting you're unmuted; learn `⌘D` / `Ctrl+D`

✓ TIPS & BEST PRACTICES

- Add a secondary time zone in Settings
- Set default event length to 25 / 50 minutes (Speedy meetings)
- Attach the agenda doc to the invite

QUICK SUMMARY

- `c t d w m` — Calendar keys
- `Find a time` — free / busy
- `Appointment schedule` — booking page
- `⌘D / ⌘E` — Meet mute / camera

Forms, Keep & Tasks

Forms question types · quizzes · responses to Sheets · Keep notes, labels & reminders · Tasks from Gmail / Docs · action items

COLLECT, CAPTURE AND TRACK

Forms builds surveys, sign-ups and quizzes and sends responses straight into a **Sheet**. **Keep** holds quick notes, checklists and reminders that sync everywhere and can be dropped into Docs. **Tasks** collects to-dos from Gmail, Docs action items and Calendar in one list.

NOTES

- Forms: Responses tab › **Link to Sheets** for live response data
- Quiz mode: Settings › Make this a quiz; set answer keys and points
- Response validation (number, regex, length) per question
- Keep: label notes, add reminders, share; Keep side panel in Docs / Gmail
- Tasks: Gmail ⋯ › Add to Tasks; assigned Docs comments appear too

AUTOMATE IT

- Form → Sheet → Apps Script `onFormSubmit` trigger for confirmations or routing
- Pre-filled form links (⋯ › Get pre-filled link) for repeat submissions

BUILD A FORM THAT FEEDS A SHEET

1. Go to `forms.new`; add a title and questions
2. Pick types (short answer, dropdown, date, file upload…)
3. Turn on **Required** and response validation where needed
4. Settings: collect emails, limit to 1 response, quiz mode
5. Responses › **Link to Sheets** (new or existing sheet)
6. Send › copy link or embed; watch rows arrive in the Sheet

WHERE TO FIND IT

TASK	WHERE
New form	forms.new
Link to Sheets	Responses › Link to Sheets
Branching	⋯ › Go to section based on answer
Pre-filled link	⋯ › Get pre-filled link
Keep side panel	Docs / Gmail right panel
Add email to Tasks	Gmail toolbar › Add to Tasks

OTHER SUITES

- ⇄ **Microsoft 365** — Microsoft Forms → Excel, OneNote / Sticky Notes, To Do
- ⇄ **Apple iWork** — no Forms app; Notes and Reminders for capture

PLATFORM DIFFERENCES

- **Browser** — full Forms editor
- **Offline** — Keep and Tasks work offline; Forms needs a connection
- **Mobile** — Keep and Tasks apps; Forms fill-in works, editing is basic

QUESTION TYPES

TYPE	USE
Short answer / paragraph	Free text
Multiple choice / checkboxes	One / many options
Dropdown	Long option lists
Linear scale / rating	1–5, stars
Date / time	Pickers
File upload	To the owner's Drive

WHERE TASKS COME FROM

SOURCE	HOW
Gmail	Add to Tasks
Docs	Assigned comments / checklists
Chat spaces	Space tasks
Calendar	Tasks show on dates
Keep	Reminders (separate)

⚠ COMMON MISTAKES

- Editing the response Sheet's structure (deleting columns) and breaking the link
- File upload questions on forms for people outside your domain (they must sign in)
- Keeping to-dos in Keep, Tasks and email flags all at once

✓ TIPS & BEST PRACTICES

- Add formulas on a separate tab, not in the response columns
- Use sections and branching to keep forms short
- Pick one task system and route everything into it

QUICK SUMMARY

- `forms.new` — new form
- `Link to Sheets` — live responses
- `Keep` — quick notes and reminders
- `Tasks` — one to-do list

Macros & Apps Script Basics

RECORD, THEN READ THE CODE

Sheets can **record macros**: Extensions › Macros › Record macro, do the steps, save. The recording is saved as **Apps Script** (JavaScript) in the file, and you can give it a shortcut. Open **Extensions › Apps Script** to edit macros, write **custom functions** you can call from cells, and automate Docs, Gmail, Drive and Calendar.

NOTES

- Absolute vs relative references are chosen when you start recording
- Shortcut keys: `⌘⌥⇧1…0` / `Ctrl+Alt+Shift+1…0` (10 slots)
- Macros live in the file's bound script project (`macros.gs`)
- First run asks for authorization; review what the script can access
- Custom functions use a JSDoc `@customfunction` tag for autocomplete

A MACRO AND A CUSTOM FUNCTION

```
/** Recorded macro, tidied: bold header, freeze, resize */
function formatReport() {
  const sheet = SpreadsheetApp.getActiveSheet();
  sheet.getRange('1:1').setFontWeight('bold');
  sheet.setFrozenRows(1);
  sheet.autoResizeColumns(1, sheet.getLastColumn());
  sheet.getRange('D2:D').setNumberFormat('$#,##0.00');
}

/**
 * Adds sales tax and rounds to cents.
 * @param {number} amount  The pre-tax amount.
 * @param {number} rate    Tax rate, e.g. 0.0825.
 * @customfunction
 */
function TAXED(amount, rate) {
  return Math.round(amount * (1 + rate) * 100) / 100;
}
```

CODE EXPLAINED

- → Run functions from the editor's Run button or Extensions › Macros
- → In a cell: `=TAXED(B2, 0.0825)` ; custom functions can't change other cells
- → `getRange('D2:D')` uses A1 notation, including open ranges
- → The JSDoc comment becomes the help text shown in the cell

OTHER SUITES

- ⇄ **Microsoft 365** — Excel macro recorder writes VBA; Office Scripts write TypeScript
- ⇄ **Apple iWork** — no recorder; AppleScript and Shortcuts automate the Mac apps

PLATFORM DIFFERENCES

- ▣ **Browser** — record, edit, run
- ▣ **Offline** — scripts don't run offline
- ▣ **Mobile** — custom functions calculate; menus and macros don't run

KINDS OF SCRIPTS

KIND	STARTS FROM
Recorded macro	Extensions › Macros
Custom function	`@customfunction` + use in a cell
Custom menu	`onOpen()` adds menu items
Trigger	On edit, on form submit, time-driven
Add-on	Packaged script others install

CUSTOM FUNCTION LIMITS

LIMIT	DETAIL
Run time	~30 seconds per call
Side effects	Can't edit other cells / send mail
Inputs	Values, not ranges as objects
Speed	Slower than named functions
Return arrays	Allowed (spills)

⚠ COMMON MISTAKES

- Recording with absolute references and running the macro from a different cell
- Calling a custom function on thousands of rows one cell at a time; pass a range and return an array
- Clicking through authorization for scripts you didn't write or read

✓ TIPS & BEST PRACTICES

- Record first, then rename the function and delete extra lines
- Use named functions (sheet 20) when a formula is enough
- Read the Execution log (View › Logs) to debug

QUICK SUMMARY

- `Extensions › Macros › Record` — record steps
- `⌘⌥⇧1` — macro shortcut
- `Extensions › Apps Script` — edit code
- `@customfunction` — formula from script

Apps Script Recipes

AUTOMATE ACROSS WORKSPACE

Apps Script can read and write Sheets in bulk, add **custom menus**, react to edits with **simple triggers**, run on a schedule with **time-driven triggers**, and reach **Gmail**, **Docs**, **Drive** and **Calendar**. For no-code automation, **Workspace Flows** and **AppSheet** cover many of the same jobs.

NOTES

- Batch: `getValues()` / `setValues()` read and write 2-D arrays in one call
- `onOpen(e)` and `onEdit(e)` are simple triggers: no setup, limited permissions
- Installable triggers (clock icon in the editor) run on a schedule or form submit
- Quotas apply (emails per day, runtime per execution)
- Use `Logger.log()` or `console.log()` and the Executions page to debug

AUTOMATE IT

- Time-driven trigger: every Monday 7am, email a summary of overdue tasks
- onFormSubmit trigger: confirmation email + route to the right owner

MENU, EDIT TRIGGER, MAIL MERGE

```
function onOpen() {
  SpreadsheetApp.getUi().createMenu('Tools')
    .addItem('Flag overdue', 'flagOverdue')
    .addItem('Send reminders', 'sendReminders').addToUi();
}

function onEdit(e) {                          // timestamp column C when B changes
  if (e.range.getColumn() === 2) e.range.offset(0, 1).setValue(new Date());
}

function sendReminders() {                    // one email per row
  const rows = SpreadsheetApp.getActive().getSheetByName('Tasks')
                .getDataRange().getValues().slice(1);
  rows.filter(r => r[3] !== 'Done' && r[2] < new Date())
      .forEach(([task, email, due]) => GmailApp.sendEmail(
        email, `Overdue: ${task}`, `"${task}" was due ${due.toDateString()}.`));
}
```

CODE EXPLAINED

- → `getDataRange()` = all used cells; `.slice(1)` drops the header row
- → Destructuring `[task, email, due]` names the first three columns
- → Simple triggers can't send email; run `sendReminders` from the menu or an installable trigger
- → Template literals (backtick strings) build message text with `${…}`

OTHER SUITES

- ⇄ **Microsoft 365** — VBA (desktop) and Office Scripts + Power Automate (cloud)
- ⇄ **Apple iWork** — AppleScript / JXA and Shortcuts on the Mac only

PLATFORM DIFFERENCES

- ▣ **Browser** — editor and triggers
- ▣ **Server-side** — triggers run without the file open
- ▣ **Mobile** — custom menus don't appear

SERVICES

SERVICE	USE
`SpreadsheetApp`	Sheets
`DocumentApp / SlidesApp`	Docs / Slides
`GmailApp / MailApp`	Send and read mail
`DriveApp`	Files and folders
`CalendarApp`	Events
`UrlFetchApp`	Call web APIs

TRIGGERS

TRIGGER	RUNS
`onOpen (simple)`	File opens
`onEdit (simple)`	User edits a cell
`Time-driven`	Hourly / daily / weekly
`On form submit`	New form response
`On change`	Structure changes

⚠ COMMON MISTAKES

- Reading cells one at a time in a loop; use `getValues()` once
- Expecting `onEdit` to fire for changes made by formulas or other scripts
- Hitting email quotas with big sends; batch and check `MailApp.getRemainingDailyQuota()`

✓ TIPS & BEST PRACTICES

- Keep secrets in Script Properties, not in code
- Add a custom menu so non-technical users can run scripts
- Try Workspace Flows before writing a script for simple automations

QUICK SUMMARY

- `getValues / setValues` — batch I/O
- `onOpen` — custom menu
- `onEdit(e)` — react to edits
- `Time-driven trigger` — scheduled runs

Gemini in Workspace

AI INSIDE THE APPS

Gemini appears as a side panel (✦) in Docs, Sheets, Slides, Gmail and Drive, and as inline tools like **Help me write**. In Sheets it can build formulas, tables and charts, and the `AI()` function runs a prompt per row. Features depend on your plan, and every result is a draft to check, not a final answer.

NOTES

- Side panel: ✦ icon top right; it can reference other Drive files with `@`
- Docs: Help me write / refine, summaries, image generation
- Sheets: "Help me organize", formula explanations, `=AI("prompt", A2)`
- Gmail: summarise thread, draft replies, Help me write
- Availability varies by Workspace edition and Google One plan

PROMPTS AND THE AI FUNCTION

```
# Side panel prompts
Docs:   "Summarize this doc in 5 bullets for a manager; flag open risks"
Sheets: "Create a formula in E2 that flags orders over $1,000 on weekends"
Slides: "Draft speaker notes for this slide, 60 seconds, plain language"
Gmail:  "Summarize this thread and list who owes what by when"
Drive:  "Find the latest budget sheet and list its totals by department"

# Sheets AI function (one call per cell)
=AI("Classify this feedback as positive, neutral or negative", B2)
=AI("Extract the company name", A2)
=AI("Write a 10-word product tagline for", C2)
```

CODE EXPLAINED

- → Name the source (`@file`), the task, the format and the audience in prompts
- → `AI()` results don't recalculate automatically; refresh them when inputs change
- → Ask Gemini to explain a formula before trusting one it wrote
- → Generated text and images still need fact and rights checks

OTHER SUITES

- ⇄ **Microsoft 365** — Copilot in each app, `COPILOT()` function, Python in Excel
- ⇄ **Apple iWork** — Apple Intelligence Writing Tools; no spreadsheet AI function

PLATFORM DIFFERENCES

- ▣ **Browser** — full side panel and AI function
- ▣ **Offline** — no Gemini
- ▣ **Mobile** — Gemini in Gmail and the Gemini app; fewer editor features

PROMPT RECIPE

PART	EXAMPLE
Task	Draft a status update
Source	using @Project Plan
Audience	for leadership
Format	5 bullets, under 120 words
Constraints	include risks, no jargon

VERIFY BEFORE USING

OUTPUT	CHECK
Formulas	Hand-check a few rows
Numbers in summaries	Trace to source cells
AI() classifications	Spot-check a sample
Drafted emails	Facts, names, tone
Generated images	Accuracy, rights

⚠ COMMON MISTAKES

- Using AI() for things a normal formula does exactly
- Pasting AI summaries into reports without checking numbers
- Vague prompts like "improve this"; say what better means

✓ TIPS & BEST PRACTICES

- Keep data in clean tables with clear headers; Gemini does better
- Paste AI() results as values once checked
- See the Agentic Coding set for prompting techniques

QUICK SUMMARY

- `✦ side panel` — ask about your files
- `Help me write` — drafts
- `=AI(prompt, cell)` — per-row AI
- `Verify` — every AI result

Sheet 30

Quick Ref — Drive, Docs, Slides & Gmail

macOS / Windows · shared · Drive · Docs · Slides · Gmail · Calendar

SHARED & DRIVE (02, 05)

ACTION	MACOS	WINDOWS / CHROMEOS
Shortcut list	⌘/	Ctrl+/
Search the menus	⌥/	Alt+/
Paste without formatting	⌘⇧V	Ctrl+Shift+V
Copy / paste format	⌘⌥C / ⌘⌥V	Ctrl+Alt+C / V
Undo / redo	⌘Z / ⌘⇧Z	Ctrl+Z / Ctrl+Y
Comment	⌘⌥M	Ctrl+Alt+M
Version history	⌘⌥⇧H	Ctrl+Alt+Shift+H
Find & replace	⌘⇧H	Ctrl+H
Drive: search	/	/
Drive: add shortcut	⇧Z	Shift+Z
Drive: share selected	.	.
Drive: new Doc / Sheet / Slides	⇧T / ⇧S / ⇧P	Shift+T / S / P
Drive: rename	N	N
Drive: shortcut list	?	?

DOCS (06–09)

ACTION	MACOS	WINDOWS / CHROMEOS
Normal text	⌘⌥0	Ctrl+Alt+0
Heading 1–6	⌘⌥1–6	Ctrl+Alt+1–6
Clear formatting	⌘\	Ctrl+\
Strikethrough	⌘⇧X	Alt+Shift+5
Numbered / bulleted list	⌘⇧7 / ⌘⇧8	Ctrl+Shift+7 / 8
Center / justify	⌘⇧E / ⌘⇧J	Ctrl+Shift+E / J
Page break	⌘↵	Ctrl+Enter
Footnote	⌘⌥F	Ctrl+Alt+F
Word count	⌘⇧C	Ctrl+Shift+C
Spelling	⌘⌥X	Ctrl+Alt+X
Suggesting / editing	⌘⌥⇧X / ⌘⌥⇧Z	Ctrl+Alt+Shift+X / Z
Voice typing	⌘⇧S	Ctrl+Shift+S
Show outline	⌘⌥A ⌘⌥H	Ctrl+Alt+A Ctrl+Alt+H
Next heading	^⌘ N, H	Ctrl+Alt N, H

SLIDES (22–23)

ACTION	MACOS	WINDOWS / CHROMEOS
New slide	⌘M	Ctrl+M
Duplicate	⌘D	Ctrl+D
Group / ungroup	⌘⌥G / ⌘⌥⇧G	Ctrl+Alt+G / Ctrl+Alt+Shift+G
Forward / to front	⌘↑ / ⌘⇧↑	Ctrl+↑ / Ctrl+Shift+↑
Present	⌘↵	Ctrl+Enter
Present from start	⌘⇧↵	Ctrl+Shift+Enter
Go to slide n	n ↵	n Enter
Black / white	B / W	B / W
Laser / notes / audience	L / S / A	L / S / A
Captions	⌘⇧C	Ctrl+Shift+C

GMAIL & CALENDAR (24–25)

ACTION	MACOS	WINDOWS / CHROMEOS
Compose	c	c
Reply / all / forward	r / a / f	r / a / f
Next / previous	j / k	j / k
Archive / delete	e / #	e / #
Label / move	l / v	l / v
Snooze	b	b
Go to inbox	g i	g i
Send	⌘↵	Ctrl+Enter
Mark read / unread	⇧I / ⇧U	Shift+I / Shift+U
Select / star	x / s	x / s
Undo action	z	z
Shortcut list	?	?

DOCS NAVIGATION & REVIEW (07, 09)

ACTION	MACOS	WINDOWS / CHROMEOS
Next / prev heading	^⌘ N,H / P,H	Ctrl+Alt N,H / P,H
Next comment	^⌘ N,C	Ctrl+Alt N,C
Next footnote	^⌘ N,F	Ctrl+Alt N,F
Doc start / end	⌘↑ / ⌘↓	Ctrl+Home / End
Insert link	⌘K	Ctrl+K
Select all matching	right-click	right-click
Open comment history	⌘⌥⇧A	Ctrl+Alt+Shift+A
Toggle compact mode	⌘⇧F	Ctrl+Shift+F

CALENDAR, MEET & FORMS (25–26)

ACTION	MACOS	WINDOWS / CHROMEOS
Create event	c	c
Today	t	t
Day / week / month / year	d / w / m / y	d / w / m / y
Next / previous period	j / k	j / k
Quick add	q	q
Search	/	/
Meet: mute / camera	⌘D / ⌘E	Ctrl+D / Ctrl+E
Meet: raise hand	^⌘H	Ctrl+Alt+H
New form	forms.new	forms.new

Sheet 31

Quick Ref — Sheets Shortcuts

macOS / Windows · navigation · selection · entry · formatting · formulas · rows / sheets

NAVIGATION (10)

ACTION	MACOS	WINDOWS / CHROMEOS
Jump to data edge	⌘ + arrow	Ctrl + arrow
Go to A1	fn⌘←	Ctrl+Home
Last cell	fn⌘→	Ctrl+End
Next / prev sheet	⌥↓ / ⌥↑	Ctrl+Shift+PgDn / PgUp
List sheets	⌥⇧K	Alt+Shift+K
Search menus	⌥/	Alt+/
Find / replace	⌘F / ⌘⇧H	Ctrl+F / Ctrl+H
Hide menus	⌘⇧F	Ctrl+Shift+F

SELECTION & STRUCTURE (10)

ACTION	MACOS	WINDOWS / CHROMEOS
Extend to edge	⇧⌘ + arrow	Ctrl+Shift + arrow
Select column / row	^Space / ⇧Space	Ctrl+Space / Shift+Space
Select data / all	⌘A	Ctrl+A
Insert rows / cols	⌘⌥=	Ctrl+Alt+=
Delete rows / cols	⌘⌥-	Ctrl+Alt+-
Hide rows / columns	⌘⌥9 / ⌘⌥0	Ctrl+Alt+9 / 0
Group / ungroup	⌥⇧→ / ⌥⇧←	Alt+Shift+→ / ←
Convert to table	⌘⌥T	Ctrl+Alt+T

ENTRY & FILL (11)

ACTION	MACOS	WINDOWS / CHROMEOS
Fill down / right	⌘D / ⌘R	Ctrl+D / Ctrl+R
Fill selection	⌘↩	Ctrl+Enter
Smart Fill	⌘⇧Y	Ctrl+Shift+Y
Date / time	⌘; / ⌘⇧;	Ctrl+; / Ctrl+Shift+;
Date and time	⌘⌥⇧;	Ctrl+Alt+Shift+;
New line in cell	⌥↩	Alt+Enter
Edit cell	↩ / F2	Enter / F2
Insert link	⌘K	Ctrl+K
Paste values only	⌘⇧V	Ctrl+Shift+V
Insert note	⇧F2	Shift+F2

FORMATTING

ACTION	MACOS	WINDOWS / CHROMEOS
Bold / italic / underline	⌘B / ⌘I / ⌘U	Ctrl+B / I / U
Strikethrough	⌘⇧X	Alt+Shift+5
Currency	⌘⇧4	Ctrl+Shift+4
Percent	⌘⇧5	Ctrl+Shift+5
Decimal number	⌘⇧1	Ctrl+Shift+1
Date	⌘⇧3	Ctrl+Shift+3
Clear formatting	⌘\	Ctrl+\
Border outline	⌥⇧7	Alt+Shift+7
Align center	⌘⇧E	Ctrl+Shift+E
Paint format	roller icon	roller icon

FORMULAS (12, 16)

ACTION	MACOS	WINDOWS / CHROMEOS
Toggle $ refs	F4	F4
Show formulas	^~	Ctrl+~
ARRAYFORMULA wrap	⌘⇧↩	Ctrl+Shift+Enter
Formula help	⇧F1	Shift+F1
Collapse / expand help	F1	F1
Accept suggestion	Tab	Tab

MACROS & TOOLS (21, 27)

ACTION	MACOS	WINDOWS / CHROMEOS
Run macro 1–10	⌘⌥⇧1–0	Ctrl+Alt+Shift+1–0
Record macro	Extensions › Macros	Extensions › Macros
Explore (suggestions)	✦ / Explore button	Explore button
Comment	⌘⌥M	Ctrl+Alt+M
Version history	⌘⌥⇧H	Ctrl+Alt+Shift+H
Shortcut list	⌘/	Ctrl+/

Sheet 32

Quick Ref — Sheets Formulas

logic · lookup · arrays · QUERY · text & regex · date · import · LAMBDA

LOGIC & MATH (14)

FORMULA	DOES
`IF(test, yes, no)`	One of two values
`IFS(t1,v1, …, TRUE,else)`	First true test
`SWITCH(x, a,"A", other)`	Match a value
`IFNA(x, alt) / IFERROR`	Catch errors
`SUMIFS(sum, rng,crit, …)`	Conditional sum
`COUNTIFS(rng,crit, …)`	Conditional count
`AVERAGEIFS / MAXIFS / MINIFS`	Conditional stats
`COUNTUNIQUE(rng) / COUNTUNIQUEIFS`	Distinct count
`SUMPRODUCT(a, b)`	Σ a × b
`ROUND(x, 2)`	Round

LOOKUP (15)

FORMULA	DOES
`XLOOKUP(k, look, ret)`	Exact match
`XLOOKUP(k, look, ret, "none")`	With miss value
`XLOOKUP(k, look, ret, , -1)`	Next smaller
`XLOOKUP(k, look, ret, , 0, -1)`	Last match
`INDEX(ret, MATCH(k, look, 0))`	Classic
`VLOOKUP(k, rng, col, FALSE)`	Legacy exact
`FILTER(rng, look = k)`	All matches
`XLOOKUP(a&b, X:X&Y:Y, Z:Z)`	Two keys

ARRAYS & QUERY (16–17)

FORMULA	DOES
`ARRAYFORMULA(expr)`	Whole-column formula
`FILTER(rng, c1, c2)`	AND conditions
`SORT(rng, col, FALSE)`	Descending sort
`SORTN(rng, 5, 0, col, FALSE)`	Top 5
`UNIQUE(rng)`	Distinct
`{A:A, C:C} / {A:A; B:B}`	Side by side / stack
`QUERY(rng, "select A, sum(E) group by A", 1)`	Summary
`… where B = 'West' order by E desc`	Filter + sort
`… pivot C`	Cross-tab
`Col1, Col2`	Columns for arrays / imports

TEXT, REGEX & DATE (18)

FORMULA	DOES
`SPLIT(t, ",")`	Split to columns
`TEXTJOIN(", ", TRUE, rng)`	Join, skip blanks
`REGEXMATCH(t, "^\d+$")`	Pattern test
`REGEXEXTRACT(t, "(\d+)")`	Pull a match
`REGEXREPLACE(t, "(?i)x", "y")`	Replace, any case
`TEXT(n, "$#,##0")`	Format as text
`EOMONTH(d, 0) / EDATE(d, 3)`	Month math
`NETWORKDAYS(a, b, hol)`	Business days
`DATEDIF(a, b, "Y")`	Whole years

IMPORT & LIVE DATA (19)

FORMULA	DOES
`IMPORTRANGE(url, "Tab!A:E")`	Another sheet
`IMPORTHTML(url, "table", 1)`	Web table
`IMPORTXML(url, "//h2")`	XPath
`IMPORTDATA(url)`	CSV
`GOOGLEFINANCE("GOOG")`	Price
`GOOGLETRANSLATE(t, "en", "es")`	Translate
`IMAGE(url, 1)`	Picture in cell
`SPARKLINE(rng)`	Chart in cell (21)

LET & LAMBDA (20)

FORMULA	DOES
`LET(n, val, calc)`	Named steps
`LAMBDA(x, calc)(5)`	Inline function
`MAP(rng, LAMBDA(x, …))`	Each cell
`BYROW(rng, LAMBDA(r, …))`	Each row
`REDUCE(0, rng, LAMBDA(a,x, …))`	Accumulate
`SCAN(0, rng, LAMBDA(a,x, a+x))`	Running total
`Data › Named functions`	Save for reuse
`=AI("prompt", A2)`	Gemini per row (29)